

The Emergence of Strategic Reasoning of Large Language Models*

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Abstract

As large language models (LLMs) have demonstrated strong reasoning abilities in structured tasks (e.g., coding and mathematics), we explore whether these abilities extend to strategic multi-agent environments. We investigate strategic reasoning capabilities – the process of choosing an optimal course of action by predicting and adapting to others’ actions – of LLMs by analyzing their performance in three classical games from behavioral economics. Using hierarchical models of bounded rationality, we compare three standard LLMs (ChatGPT-4, Claude-3.5-Sonnet, Gemini 1.5) and three reasoning LLMs (OpenAI-o1, Claude-4-Sonnet-Thinking, Gemini Flash Thinking 2.0). Our results show that reasoning LLMs exhibit more sophisticated strategic reasoning than standard LLMs, which do not demonstrate substantial strategic reasoning capabilities. Comparisons with human behavior suggest that LLM strategic reasoning is task-dependent rather than uniformly superior or inferior to human reasoning. Nonetheless, reasoning LLMs frequently match or exceed human performance, representing a significant shift in the strategic reasoning capabilities exhibited by LLMs.

Keywords: Strategic Reasoning, Large Language Models, Level- k Model, Cognitive Hierarchy Model

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1 Introduction

As the reliance on Large Language Models (LLMs) is now pervasive, it is vital to assess whether the mechanisms through which LLMs reason are also capable of *strategic* reasoning – the process of choosing an optimal course of action by predicting and adapting to the actions of others in multi-agent environments. While research has shown LLMs’ reasoning skills in individual decision-making (economic rationality) (Chen et al., 2023; Kim et al., 2024), mathematics (Ahn et al., 2024; Zhou et al., 2024), simple reasoning problems (Webb, Holyoak and Lu, 2023; Hagendorff, Fabi and Kosinski, 2023), and coding tasks (OpenAI, 2024; Anthropic, 2024), this does not inherently imply they possess strategic reasoning capabilities. In fact, strategic reasoning appears to be an emergent capability that is largely absent in earlier standard LLMs and becomes clearly observable only with the advent of reasoning LLMs. This represents a major step forward in strategic reasoning capabilities since the development of LLMs.¹

Such capabilities are crucial for the increasingly popular use of Agentic AI (AAI), which involves multi-agent decision-making environments with AI/LLM systems having competitive or cooperative objectives that require constant anticipation and adaptation for autonomous, real-time tasks. Key real-world applications of AAI include autonomous financial agents that anticipate market movements to make trading decisions (Luo and Duan, 2025; An et al., 2024), supply chain systems that coordinate resources by predicting demand (Almutairi and Kim, 2025; Xu et al., 2024), and negotiation systems requiring agents to infer and strategically anticipate responses (Abdelnabi et al., 2024; Kwon et al., 2024). Unlike individual decision-making, these types of applications require agents to form beliefs about how others (LLMs and/or people) will act in order to complete tasks.² The strategic sophistication of these AI agents therefore has consequences for human decision-makers. As people increasingly use LLMs for strategic advice (negotiations, investments, etc.), more sophisticated LLMs may provide better outputs for improved decision-making in strategic contexts. From a market perspective, variations in strategic sophistication across LLMs create potential asymmetries where users of less sophisticated LLMs may be exploitable by those employing more strategically sophisticated systems. Thus, investigating the strategic sophistication of LLMs has broader implications beyond benchmarking model performance.

¹While earlier standard LLMs trained before the first reasoning LLMs demonstrate a lack of inherent strategic reasoning, standard LLMs developed later are not incapable of strategic reasoning. After the introduction of reasoning LLMs, standard LLMs are likely to be exposed to more reasoning-based thinking and tasks (or may even have benefited from fine-tuning or knowledge distillation from reasoning LLMs), allowing standard LLMs to develop higher-order strategic reasoning in a way that is crudely analogous to reasoning LLMs.

²More generally, achieving Artificial General Intelligence (AGI) - AI that performs a wide range of cognitive tasks at, or beyond, human-level proficiency (Singh et al., 2024) - necessitates systems capable of strategic reasoning, making it essential to benchmark these capabilities (Mumuni and Mumuni, 2025).

In this study, our objective is to explore this emergence of strategic reasoning capabilities (Wei et al., 2022b; Schaeffer, Miranda and Koyejo, 2023) leveraging OpenAI’s ChatGPT-4 and OpenAI-o1-preview (GPT-4, GPT-o1), Anthropic’s Claude-3.5-Sonnet and Claude-4-Sonnet-Thinking (Cl-3S, Cl-4ST), and Google’s Gemini 1.5 Flash and Gemini Flash Thinking 2.0 Experimental (Ge-1.5, Ge-2T). We investigate whether reasoning LLMs, specifically designed with reasoning techniques (GPT-o1, Cl-4ST, and Ge-2T), exhibit sophisticated strategic reasoning abilities, and whether such abilities were already present in more standard LLMs.³ To evaluate and deepen our understanding of LLMs’ strategic reasoning, we adopt a behavioral economics framework – a ubiquitous approach in studying human behavior – to provide valuable insights into how well LLMs can navigate strategic environments.⁴

We explore strategic reasoning using three classical games from the behavioral economics literature: the p-Beauty Contest Game (pBCG), the Guessing Game (GG), and the 11-20 Money Request Game (MRG) (Nagel, 1995b; Costa-Gomes and Crawford, 2006; Arad and Rubinstein, 2012, respectively), allowing us to evaluate LLMs by estimating their depth of strategic reasoning and investigate whether we observe similarities/differences across games.^{5,6} These games span distinct reasoning demands. In the pBCG, iterated best-response reasoning leads to a unique symmetric equilibrium, so equilibrium actions are identical between players. By contrast, the GG features role-specific incentives and yields asymmetric equilibrium actions between players. Finally, the MRG requires probabilistic reasoning: an equilibrium is a randomization of actions rather than a single deterministic action for players. Thus, differences in the equilibrium form across the three games require distinct depths of reasoning to obtain the equilibrium. Using such games provides us with essential tools to evaluate LLMs, as the focus is on logic and strategic reasoning through clear, well-defined rules and, vitally, action spaces that allow us to directly test the depth of strategic reasoning abilities.

Two of the most profound models from behavioral economics that capture the recursive

³It should be noted that although reasoning LLMs are trained in chain-of-thought (CoT), they represent more than just standard LLMs with CoT prompts. Reasoning LLMs incorporate fundamental architectural and training differences beyond prompting alone. For instance, they use reinforcement learning from human feedback (RLHF) optimized specifically for long chain-of-thought (CoT) trajectories, inference-time scaling with extended *thinking* budgets (e.g., generating thousands of internal reasoning tokens before final output), and self-correction mechanisms during reasoning traces that mimic search algorithms rather than simple next-token prediction. See the detailed discussion in Yeo et al. (2025).

⁴A growing literature studies the strategic capabilities of LLMs (Duan et al., 2024a; Costarelli et al., 2024; Jia et al., 2025; Zhang et al., 2025b). These studies primarily benchmark LLM performance across multiple games under varying prompting designs. In contrast, we hold the prompt fixed and focus on a head-to-head comparison between reasoning-focused and standard LLMs. Moreover, our set of games is selected to span a range of strategic reasoning demands.

⁵In human subjects, Georganas, Healy and Weber (2015) show that strategic reasoning can differ across games.

⁶Using behavioral games (dictator game, ultimatum game, trust game, bomb risk game, and public goods game), Mei et al. (2024) show that LLMs exhibit behavioral traits such as trust, fairness, risk-aversion, altruism, and cooperation, with responses that are statistically indistinguishable from human behavior.

process involved in limited strategic sophistication are the level- k model (Stahl and Wilson, 1994, 1995; Nagel, 1995b) and the cognitive hierarchy (CH) model (Camerer, Ho and Chong, 2004).⁷ These alternative models of bounded rationality have been able to provide more accurate predictions of behavior. Studies in behavioral economics have largely focused on human adults with some research done with children (Brocas and Carrillo, 2020; Hermes and Schunk, 2022; Crawford, Costa-Gomes and Iriberri, 2013). We apply both models to evaluate LLMs by measuring their level of strategic reasoning represented by a hierarchy of iterated reasoning – an investigation that has not been conducted for LLMs.

The *lowest rank* in the hierarchy, say rank-0, is completely non-strategic, that is, actions are made without taking into account the actions of others.⁸ The next rank represents more strategic behavior (rank-1) with beliefs that all others are non-strategic (rank-0). Similarly, rank-2 beliefs are that all others consist of lower ranks. In the level- k model, these lower ranks, referred to as levels, are always exactly one level lower in the reasoning hierarchy, e.g., level-2 (L_2) assumes that all others are level-1 (L_1). In the CH model, these lower ranks, referred to as steps, accommodate a combination of all lower steps of reasoning, e.g., step-2 assumes that all others are a mixture of step-0 and step-1. As we move up the hierarchy, this pattern continues for higher ranks. In this process, knowing how many stages of iterated reasoning the others are utilizing (i.e., the others’ levels) is essential. In this way, we aim to answer to what extent are LLMs strategically sophisticated.

To evaluate the strategic reasoning of LLMs, we instruct LLMs to play the different games under various treatments. To obtain a distribution of responses from each LLM, we request 100 independent responses under a fixed prompt and temperature, allowing us to approximate the model’s conditional choice distribution. Additional treatments include varying the parameters of the games themselves.⁹ By estimating the level- k model and the CH model, we are able to determine the depth of reasoning (up to a fixed level/step) by matching the theoretical predictions of each model for each game with the observed data. Our overall results are (i) reasoning LLMs demonstrate higher strategic reasoning compared to standard LLMs, (ii) with sufficient feedback, LLMs are able to learn and achieve higher orders of strategic reasoning (especially standard LLMs), (iii) reasoning LLMs learn faster than standard LLMs, and (iv) standard LLMs consistently exhibit worse strategic reasoning

⁷Abundant studies in behavioral economics have shown that the Nash equilibrium, derived from the assumption of full rationality, does not match well with observed play (Camerer, Loewenstein and Rabin, 2004; Camerer, 2011; Bernheim, DellaVigna and Laibson, 2019). This has given rise to models of limited strategic reasoning.

⁸A common assumption is that rank-0 behavior is consistent with choosing an action as if drawing uniformly from the action space.

⁹The temperature parameter of LLMs can also influence output in terms of randomness and creativity. We use a baseline temperature of 0.5 and also test a high (0.75) and low (0.25) temperature. Section B.5 of Appendix B contains the results of the high and low temperatures with the overarching takeaway that temperature has little effect on observed strategic reasoning.

than typical human subjects (from the experimental economics literature).

The enhanced reasoning techniques of the reasoning LLMs are the most likely explanation for their higher strategic reasoning. Specifically, GPT-o1, Cl-4ST, and Ge-2T leverage specialized large-scale reinforcement learning to enhance their reasoning skills, with chain-of-thought reasoning – a process where the model tackles tasks step-by-step, approaching problems incrementally (Wei et al., 2022a), which, in practice, means training LLMs to break down complex tasks into smaller, manageable steps, mirroring human-like reasoning.¹⁰ With this approach, reasoning LLMs refine their reasoning by exploring various strategies and recognizing potential mistakes, further enhancing their problem-solving capabilities in an iterative and (seemingly) reflective process, ensuring more careful evaluation before responding.¹¹

Using standard methodologies and models from behavioral and experimental economics, we are able to concretely evaluate the strategic reasoning capabilities of LLMs using models of bounded rationality. We contribute to the ongoing interest in comparing reasoning and standard LLMs and in comparing LLMs with human behavior (Latif et al., 2025; Zhang et al., 2025a). Further, we highlight that with sufficient feedback, even LLMs which exhibited poor performance in one-shot games can refine their strategies, which draws a parallel with human subjects (Nagel, 1995b; Ho, Camerer and Weigelt, 1998; Weber, 2003) and the reasoning LLMs. Incorporating thinking/reasoning skills into the mechanism through which LLMs operate provides a clear inherent indication as to why reasoning LLMs perform substantially better than standard LLMs and are capable of outperforming human subjects in many of these classic games, though important limitations remain depending on game structures and/or parameters. Our study emphasizes the importance of continuous improvement in these mechanisms, particularly for interactive tasks requiring higher-order reasoning.

2 Methods

Collecting LLM data

We collect data from LLMs from OpenAI, Anthropic, and Google using their APIs. The exact model names of each LLM used are: gpt-4, o1-preview-2024-09-12, claude-3-5-sonnet-20241022, claude-sonnet-4-20250514, gemini-1.5-flash, gemini-2.0-flash-thinking-exp-1219. For consistency and replicability, we used the same system message across all games, with differences only in the number of rounds played. We tell LLMs to state their final answer in [] to allow easy extraction of their final response. The rules and description of the games were provided in the assistant message. The full set of prompts is contained in Appendix B.

Human subject data

¹⁰As opposed to just prompting them to use chain-of-thought reasoning.

¹¹OpenAI-o1-preview is the first model of this type and therefore serves as an essential benchmark.

We use human subject data from seminal papers in behavioral economics to compare with LLMs. For the pBCG, we use data from the seminal papers, as well as other published papers for the alternative conditions (Nagel, 1995*b*; Hermes and Schunk, 2022; Bosch-Domènech et al., 2002). For the GG, we use data from the original paper (Costa-Gomes and Crawford, 2006). For the MRG, we use data from the original paper (Arad and Rubinstein, 2012). These serve as essential benchmarks for comparison with LLM performance; their results are representative of human subject performance. As mentioned, all datasets come from published work and were subject to their own ethics committees.

Estimation. Based on the theoretical predictions, we estimate the parameters for both the level- k and CH models by maximum likelihood estimation (MLE).¹² For the level- k model, we estimate the proportions of responses corresponding to each level from L_0 , L_1 , L_2 , L_3 , L_4 , and L_∞ (Nash) predictions. We assume that each response is a best guess with (independent and identically distributed) noise. Estimation of the level- k model gives the distribution of responses according to the different levels of reasoning. For example, if responses are concentrated around the lower-level types’ guesses, compared to higher-level types’ guesses, then overall responses from the former must exhibit less strategic reasoning than the latter. The estimated level- k model will reflect this in estimating higher frequencies for the lower levels of reasoning. Similarly for the CH model, we estimate the distribution of step-types using a Poisson distribution (fully parameterized by τ), as defined above, with a similarly defined noisy best guess based on the (conditional) distribution of lower step-types. For example, a higher/lower estimated τ is indicative of higher/lower level reasoning, on average, as this indicates more/fewer (average) steps of reasoning.¹³ As the CH model imposes a one-parameter Poisson distribution over reasoning steps, we interpret τ jointly with the more flexible level- k estimates rather than as a standalone measure of model capacity.¹⁴ A detailed description of the estimation for both models for each game is provided in Sections B.4.1 to

¹²Using standard MLE techniques, we estimate the level- k model and the CH model. Stata estimation codes for these models can be found in Moffatt (2015), a textbook for estimation in behavioral and experimental economics. We translated and edited the code from Stata to R; we obtain the same results from practice datasets from Moffatt (2015) to ensure replicability. Technical estimation details can be found in Appendix B. Further, we do distributional tests (KS tests) to compare raw responses. We use both one-sided tests to compare rationality using the first-order stochastic dominance and two-sided tests to compare the distribution of responses.

¹³For example, a $\hat{\tau} = 0$ implies that all players are step-0 types, whereas $\hat{\tau} \rightarrow \infty$ indicates players are fully rational, as *infinitely* iterated elimination of dominated strategies leads to a unique Nash equilibrium. A $\hat{\tau} = 1.5$ indicates that the average step-type is 1.5. Further, as the frequency of types $f(k)$ is Poisson-distributed, the ratio $\frac{f(k-1)}{f(k-2)}$ favors step- $(k-1)$ types when τ is large, i.e., step- k types play as if there are predominantly step- $(k-1)$ types when τ is large.

¹⁴This Poisson structure follows the behavioral economics literature on human strategic reasoning and is used here as a parsimonious behavioral benchmark, not as a claim about LLM architecture. LLM response generation need not imply a Poisson-distributed hierarchy of reasoning steps. When responses are multimodal, the best-fitting CH model may produce a relatively low τ even if the level- k estimates reveal nontrivial higher-level responses.

B.4.3 of Appendix B.

This estimation strategy requires an important interpretive qualification. Cross-sectional variation in human subject data reflects heterogeneity in beliefs, preferences, cognitive sophistication, and decision noise. By contrast, under a fixed prompt and temperature, responses collected from LLMs display variation across repeated queries because of stochastic decoding. Thus, the 100 LLM responses should not be interpreted as equivalent to responses from 100 independent human subjects, but as an approximation of each model’s conditional choice distribution in a given strategic environment. The level- k and CH model estimates summarize the strategic sophistication reflected in this stochastic response policy, with human-subject data serving as behavioral benchmarks.

3 p-Beauty Contest Game

We begin with an overview of the pBCG, examining its theoretical predictions from behavioral game theory, which accounts for both full and bounded rationality. After briefly describing how these models are estimated, we explain our approach to implementing the pBCG with LLMs.

Game. A group of $n \geq 2$ players simultaneously choose a number in the interval $[0, 100]$. The winner of the game is the player whose number is closest to a given target multiplier (p) of the average number chosen by all players.

Theoretical Predictions. Nash equilibrium requires that no player have an incentive to deviate from a given strategy profile, assuming all other players’ strategies remain the same. The equilibrium is based on the assumption of common knowledge of rationality: all players are rational, all know that all are rational, and so on. In the pBCG with $p < 1$, a unique Nash equilibrium is that all players choose 0 as no player gains by choosing another number when all others choose 0. Similarly, in the pBCG with $p > 1$, a unique Nash equilibrium is that all players choose 100.

Both in the level- k and the cognitive hierarchy (CH) models, it is assumed that players are boundedly rational in the sense that each player believes their level/step of reasoning is higher than that of others. Specifically, the level- k model assumes that a player believes that all others are one level lower in their reasoning compared to themselves. The CH model assumes that a player believes that the reasoning steps of the others are distributed (strictly) below her own, according to a Poisson distribution. For both models, we use the concept of *best guess* which is a target number based on an average guess of numbers from other players

and the target multiplier (p) .¹⁵

In the level- k model, the best guess for each reasoning level, denoted by l_k for $k = 0, 1, \dots$, is defined as:

$$l_0 = 50, \quad l_1 = l_0 \times p = 50p, \quad l_2 = l_1 \times p = 50p^2, \dots, \quad l_k = l_{k-1} \times p = 50p^k,$$

where an L_0 type chooses $l_0 = 50$, which is the average guess when choosing uniformly at random from $[0, 100]$. An L_k type, $k = 1, 2, \dots$, believes that all other players are L_{k-1} , choosing $l_{k-1} = 50 \times p^{k-1}$. Clearly, l_k converges to 0 and 100 for $p < 1$ and $p > 1$, respectively, in the interval $[0, 100]$ when the reasoning level k goes to infinity.

The CH model assumes that each player is playing against a distribution of players who are using lower reasoning steps. This distribution is given by a Poisson distribution $f(\cdot; \tau)$, where τ is the mean (and the variance) of the number of reasoning steps. Specifically, a step- k type believes that they are playing against a distribution of players with reasoning steps from 0 to $k - 1$. This distribution is given by $f_k(j; \tau)$, $j = 0, 1, \dots, k - 1$, such that

$$f_k(j; \tau) = \left(\frac{e^{-\tau} \tau^j}{j!} \right) / \left(\sum_{m=0}^{k-1} \frac{e^{-\tau} \tau^m}{m!} \right), \quad (1)$$

where $f_k(j; \tau)$ is the conditional probability for $j = 0, 1, \dots, k - 1$.

The best guess for reasoning steps, denoted by s_k for $k = 1, 2, \dots$, is defined as

$$s_k = p \times \left(\sum_{j=0}^{k-1} f_k(j; \tau) s_j \right),$$

where the average guess for step-0 is $s_0 = 50$, that is, a random choice from the interval $[0, 100]$. A step- k type believes that all others are step-0 to step- $(k - 1)$, according to the Poisson probability mass function $f_k(j; \tau)$. For example, with $\tau = 1.5$, a step-2 player believes that they are playing against a combination of step-0 and step-1 players, with the proportions of each being 40% and 60%, respectively. Given these proportions, a step-2 player would choose to play p times the expected response of the other players.

3.1 Design

The baseline condition refers to a specific set of parameters of the pBCG.

¹⁵In game theory, the concept of a *best response* is often employed to derive an equilibrium – a set of strategies that provide the most favorable outcome based on a belief about the strategies of others. In games such as these in which a best response given a belief may not be uniquely defined, we use a best guess as a reasonable best response, common to the behavioral economics literature, making it possible to identify a level/step of reasoning.

Baseline condition: The total number of participants is 11 ($n = 11$), the target statistic is the average, and the target multiplier is $2/3$ ($p = 2/3$). This represents a classical pBCG with the most commonly used parameters in the literature. We ask each LLM to complete the baseline pBCG 100 times, providing us with a distribution of responses.

Alternative conditions: To provide comparisons with the baseline condition, we change one of the parameters while keeping the others fixed. LLM responses are collected for (i) $p \in \{1/2, 4/3\}$, (ii) $n \in \{2, \text{unspecified}\}$, (iii) the target statistic is the median (instead of average). Varying these parameters does not change the substance of the baseline pBCG, but offers additional insights into LLM strategic reasoning.

Compared to the baseline condition ($p = 2/3$), we expect LLM responses to be lower on average when $p = 1/2$, since predictions at all levels/steps of reasoning are lower under $p = 1/2$, even though the rational prediction coincides across the two cases. When $p = 4/3$, the rational prediction changes to 100, whereas it is 0 in the baseline condition. In addition, when $n = 2$ (with $p = 2/3$), the pBCG reduces to a setting in which the lower number always wins, which is not the case in the baseline condition ($n = 11$). For the remaining alternative conditions, we do not have distinct theoretical implications. Instead, we compare LLM responses with known evidence documented in the literature.

Multiple rounds: We collect responses from a repeated version of the pBCG, with feedback, conducted over 10 rounds, using the same parameters as the baseline condition ($n = 11$), including $p = 4/3$. Each of the LLMs plays within a group of 10 other LLM subjects of the same model. After each round, each player receives information about the average bid and the target number ($p \times \text{average}$), where *average* is computed as the mean of the responses generated independently by the 11 LLM subjects in the previous round. Additionally, players are privately informed of their win (or loss) but not the identity of the winner nor the winning bid (following Ho, Camerer and Weigelt (1998)). Players are provided with the cumulative history of the game from round 1 to 10 within their prompts.

3.2 Baseline Condition Results

Before proceeding with data collection, we conducted a series of preliminary checks involving several questions assessing whether LLMs understand the game-specific rules and winning conditions – and whether they choose a best response when given the actions of the other players to confirm their ability to respond optimally. We asked each LLM to answer each question 25 times. Each LLM consistently answered all questions about the rules and correctly identified whether the winning conditions were satisfied. All LLMs also provided appropriate answers for a best response in the baseline condition.

The reasoning LLMs are more strategically sophisticated than the standard LLMs by at

least one to two levels of thinking on average, with GPT-o1 exhibiting the highest level of strategic reasoning overall, summarized in Figure 1.¹⁶ As is well known in the experimental economics literature (see Nagel, 1995*b*; Bosch-Domènech et al., 2002, as representative examples), human subjects are typically a combination of L_0 , L_1 , L_2 , and L_∞ , with L_∞ types being rare. To draw direct comparisons with human subjects, we also do estimation using data from Nagel (1995*b*) (different p 's), Grosskopf and Nagel (2008) ($n = 2$), Bosch-Domènech et al. (2002) ($n = \text{unspecified}$), and Hermes and Schunk (2022) (median). From Figure 1, we observe that the distribution of levels for GPT-o1, Cl-4ST, Ge-2T, and Cl-3S is shifted toward higher levels of strategic reasoning compared to human subjects. Overall, GPT-o1 and Cl-4ST exhibit the highest levels of strategic reasoning by large margins.¹⁷

Table 1 shows estimates for τ alongside respective 95% confidence intervals.¹⁸ On average, Ge-1.5 exhibits just under one step of strategic reasoning with estimates not dissimilar to human subjects (see Table A1). GPT-o1 and Cl-4ST vastly outperformed the other LLMs, and exhibit higher levels of strategic reasoning than human subjects, with GPT-o1 marginally exhibiting the highest average steps of reasoning, consistent with the estimates from the level- k model.

3.3 Alternative Conditions Results

As with the baseline condition, we check whether LLMs understand the game (specific rules & win condition, and best responses). All understanding questions were answered correctly by each LLM.

From Figure 1, Ge-1.5 is consistent with its baseline estimates and the $n = \text{unspecified}$ condition but shows a lack of strategic reasoning in the remaining conditions, exhibiting overall drops in strategic reasoning from baseline. Cl-3S only manages to stay at L_2 for the median condition with drops to combinations of L_0/L_1 and small proportions of L_2 behavior. Only GPT-o1 and (marginally) Ge-2T exhibit L_∞ behavior. This is particularly true when $n = 2$, where only GPT-o1 is virtually consistent with a fully rational player.¹⁹ Overall, for all conditions except when $p = 4/3$, only the reasoning LLMs exhibit significant L_2+ behavior, although with Cl-4ST showing significant drops compared to baseline, and with

¹⁶All non-zero proportions in Figure 1 are statistically significant at the 5% level with inference based on randomized resampling (bootstrap). For future reference, this is also true for Figures 3 and 4.

¹⁷Using data from Nagel (1995*a*) ($N = 67$), we also conduct Kolmogorov-Smirnov (KS) tests for the distribution of raw results. Results are summarized in Table A1. Focusing on those cases in which exactly one of the one-sided tests is rejected, one-sided KS tests suggest that GPT-4, GPT-o1, Cl-4ST, and Ge-2T outperform human subjects at any reasonable significance level. Two-sided tests suggest the distributions between LLMs and human subjects are all statistically different from each other at any reasonable significance level. Data made available at (Nagel et al., 2002).

¹⁸Confidence intervals are estimated from randomized resampling (bootstrap). This is true for all tables with estimates from the CH model in this paper.

¹⁹Increases in strategic reasoning are not surprising in this condition, given that the game is less complicated when there are fewer players, but with dramatic/modest improvements for GPT-o1/Ge-2T.

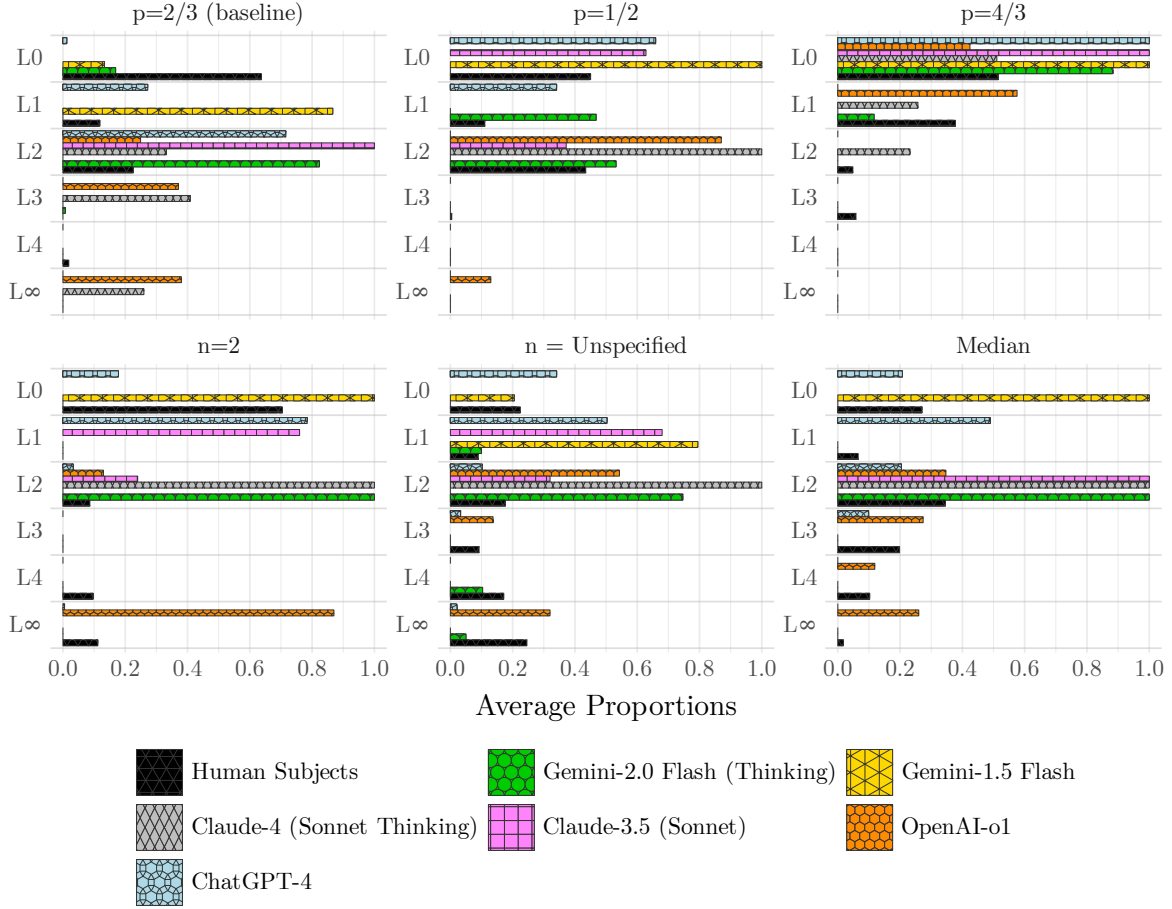


Figure 1: Estimates of the proportions of level- k types for the pBCG

GPT-o1 exhibiting the highest overall strategic reasoning.

When $p = 4/3$, only the reasoning LLMs exhibit any non-random behavior. Despite this, compared to their baselines, reasoning LLMs drop drastically in their observed strategic reasoning. A plausible explanation for poor performance in this condition is that LLMs may have been extensively exposed to pBCGs which involve iterating downward to a lower number (i.e., every other condition) as the literature mainly focuses on these versions of pBCG. This lack of flexibility means it may be difficult to adapt to a condition which involves a different iterative process, even when the difference may appear relatively minor to human decision-makers.

When compared with human subjects, the standard LLMs are typically dominated in strategic reasoning for all the alternative conditions. Cl-3S performs comparably to human subjects when $p = 1/2$ but is dominated for all other conditions. Both Cl-4ST and Ge-2T perform better and are comparable or better in all conditions except the $n = \text{unspecified}$

	$p = 2/3$ (baseline)	$p = 1/2$	$p = 4/3$	$n = 2$	$n =$ unspecified	Median
GPT-4	2.39 (2.23,2.46)	0.56 (0.56,0.79)	0.00 (0.00,0.00)	0.87 (0.79,0.94)	0.83 (0.69,1.01)	1.09 (0.83,1.36)
GPT-o1	4.42 (4.00,4.77)	2.52 (2.52,2.99)	0.95 (0.74,0.96)	7.09 (6.42,8.00)	3.43 (3.11,3.76)	4.00 (0.00,4.86)
Cl-3S	2.87 (2.87,2.87)	1.50 (1.31,1.57)	0.00 (0.00,0.00)	1.17 (1.12,1.24)	1.26 (1.18,1.34)	2.87 (2.65,2.88)
Cl-4ST	4.00 (0.00,4.23)	2.83 (2.70,2.99)	0.00 (0.00,0.00)	2.33 (2.24,2.42)	2.15 (2.10,2.21)	2.45 (2.39,2.54)
Ge-1.5	0.86 (0.79,0.91)	0.00 (0.00,0.00)	0.00 (0.00,0.00)	0.00 (0.00,0.00)	0.77 (0.70,0.84)	0.00 (0.00,0.00)
Ge-2T	2.46 (2.39,2.72)	2.12 (1.99,2.12)	0.00 (0.00,0.00)	1.86 (1.79,1.91)	2.22 (2.01,2.41)	2.46 (2.46,2.92)
Human Subjects	0.00 (0.00,1.09)	1.08 (0.00,2.14)	0.36 (0.00,0.57)	0.00 (0.00,0.00)	3.01 (2.91,3.12)	2.46 (0.00,2.46)

Table 1: Estimates of τ from the CH model for different pBCG parameters. 95% confidence intervals (bootstrapped) are shown in parentheses.

condition. GPT-o1 outperforms human subjects in terms of strategic reasoning in all alternative conditions, except when $p = 4/3$ in which case it is slightly outperformed by human subjects, yet still comparable in reasoning.²⁰

The estimates of the CH model corroborate the finding of low strategic reasoning across all alternative conditions for standard LLMs and that all reasoning LLMs (weakly) outperform standard LLMs, summarized by Table 1. When $p = 4/3$, all LLMs perform poorly, with GPT-o1 barely outperforming the other LLMs.²¹ From Table 1 we also observe that human subjects overall outperform standard LLMs in strategic reasoning, except when $n = 2$ where human subjects perform surprisingly poorly, as discussed in Grosskopf and Nagel (2008). Cl-3S is comparable with human subjects in most conditions except the $n =$ unspecified condition, in which human subjects exhibit almost 2 more steps of reasoning. Ge-2T is comparable in the median condition, and outperforms human subjects when $p = 1/2$, but is outperformed otherwise. GPT-o1 substantially outperforms human subjects in all treatments in terms of strategic reasoning, by almost 2 steps of thinking on average, with only the $p = 4/3$ condition being comparable with human subjects.

²⁰We conduct similar KS tests to those conducted for the baseline condition to compare raw responses. Results are summarized in Table A1. We focus on cases in which exactly one of the one-sided tests is rejected. For the target proportion conditions ($N_{p=1/2} = 48$; $N_{p=4/3} = 51$), all LLMs, excluding Cl-4ST, are outperformed by human subjects with Ge-2T outperforming when $p = 1/2$ (Nagel, 1995b). GPT-o1 outperforms when the median ($N = 120$) and $n =$ unspecified ($N = 1468$) conditions (Hermes and Schunk, 2022; Bosch-Domènech et al., 2002). For the $n = 2$ alternative condition in (Grosskopf and Nagel, 2008) ($N = 132$), only GPT-o1 clearly outperforms human subjects.

²¹This is consistent with Ho, Camerer and Weigelt (1998) who also estimate $\tau = 0$ when $p = 1.3$ (> 1).

3.4 Multiple Rounds Results

From the previous analysis, it is evident that the majority of LLMs display limited strategic reasoning in one-shot pBCG, particularly the standard LLMs. To investigate further, we explore a repeated version of pBCG with multiple rounds to see whether LLMs can learn to adjust their responses.

From the top layer of figures in Figure 2, across all LLMs, GPT-o1’s initial responses are always closest to the fully rational prediction, approximately exhibiting L_2 behavior. For $p = 2/3$, the initial responses of the other LLMs are consistent with L_0 and L_1 (or approximate L_2 behavior), broadly consistent with their one-shot levels of strategic reasoning. A similar result occurs when $p = 4/3$ with only L_0 behavior being observed.

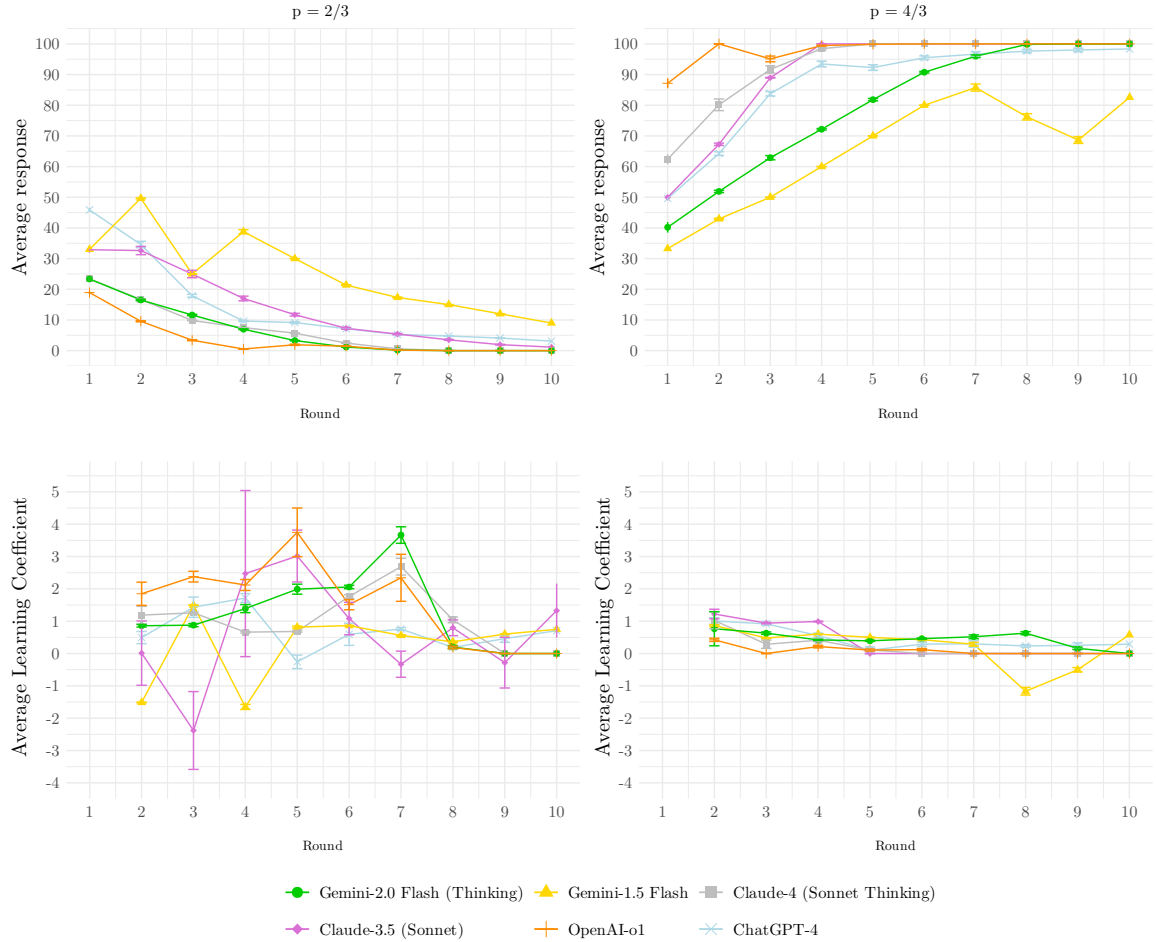


Figure 2: Time series of responses and learning coefficients for the repeated pBCG. Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals.

When $p = 2/3$, all LLMs exhibit a downward trend to the rational prediction; Ge-1.5 only

manages to converge to approximately L_3 behavior, but does not approximate L_∞ behavior unlike the other LLMs which do get to (in the close vicinity of) the rational prediction by round 10. Likewise, for $p = 4/3$, all LLMs show upward trends across all rounds with only Ge-1.5 not achieving the rational prediction. In general, the reasoning LLMs reach the rational prediction more quickly than the standard LLMs that eventually converge. Nevertheless, the convergence observed in both reasoning and standard LLMs indicates that they can learn and adapt to strategic environments when provided with sufficient feedback. Among all models, GPT-o1 performs best, exhibiting both the fastest convergence and the initial responses closest to the rational prediction.

To understand how the rate of learning differs across LLMs, we calculate a round-by-round adjustment coefficient $\alpha_{i,r}$ for LLM subject i . For each LLM subject i in round r , let $g_{i,r}$ denote the guess and let w_{r-1} denote the previous round’s winning target which is p multiplied by the group average guess in round $r - 1$. We define

$$\alpha_{i,r} = \frac{g_{i,r} - g_{i,r-1}}{w_{r-1} - g_{i,r-1}}.$$

Thus, $\alpha_{i,r} = 1$ corresponds to adapting exactly to the previous round’s target, whereas $\alpha_{i,r} \in (0, 1)$ indicates partial adjustment. Values greater than one $\alpha_{i,r} > 1$ capture strategic anticipation of the moving target, while $\alpha_{i,r} < 0$ indicates active divergence from feedback.^{22,23}

From the bottom layer of Figure 2, for $p = 2/3$, reasoning LLMs display higher learning coefficients ($\alpha_{i,r} > 1$), particularly in Rounds 2 to 5, showing forward-looking anticipation of the falling target before cleanly dropping to $\alpha_{i,r} = 0$ once the Nash equilibrium is achieved. Conversely, standard LLMs exhibit sluggish adjustments, even showing some negative adaptations (increasing average responses). By contrast, for $p = 4/3$, as reasoning LLMs tend to the Nash equilibrium earlier, their subsequent adjustment essentially drops to zero for the remainder of the rounds. Although the standard LLMs converge more slowly (if at all), this implies a more steady, positive adjustment across rounds as they still effectively continue to

²²When an LLM subject perfectly matches the prior target, their adaptation for that step is considered complete, and we set $\alpha_{i,r} = 0$.

²³We thank the referee for this suggestion.

learn.^{24,25}

We are able to draw comparisons and identify several similarities with human subjects in Ho, Camerer and Weigelt (1998) and Weber (2003). For human subjects, the rate of convergence (when convergence occurs) is much faster when $p = 4/3$ compared to $p = 2/3$.²⁶ In general, for the reasoning LLMs, the patterns of convergence and the (boundedly rational) responses in the first round are comparable with human subjects in Ho, Camerer and Weigelt (1998) when $p = 2/3$. When $p = 4/3$, LLMs seem to converge to L_∞ behavior faster than human subjects. Overall, from their observed behavior, LLMs are capable of learning and adapting in ways that are comparable to human subjects. Only the reasoning LLMs perform consistently better than human subjects, a result which is consistent throughout the entire analysis of pBCG.

4 Guessing Game

The Guessing Game (GG) from Costa-Gomes and Crawford (2006) is a two-player game in which both players make a guess about the opponent’s guess, similar to the pBCG. Unlike the pBCG, both players may have different target multipliers and different domains for their answers, which is common knowledge to both players.²⁷ This asymmetry in target multipliers and answer domains leads to an asymmetric rational prediction in Nash equilibrium between both players. An appealing feature of the GG is its dominance-solvability within a finite number of reasoning iterations.²⁸ This can bring an additional advantage as we can easily identify subjects’ decision rules, and therefore, types, for both the level- k and CH models,

²⁴We conduct one-sided Wilcoxon rank-sum tests to confirm our analyses. First, we compare pooled learning performance, i.e., reasoning LLMs overall vs. standard LLMs overall. When $p = 2/3$, the hypothesis that reasoning LLMs learn faster is well supported ($p < 0.01$), with the reasoning LLMs averaging $\bar{\alpha} = 1.27$ compared to the standard group’s $\bar{\alpha} = 0.52$. When $p = 4/3$, the aggregate test statistically captures the earlier convergence of reasoning LLMs (0.25 vs. 0.34, $p \gg 0.10$) as previously discussed. We also compare each reasoning LLM with each standard LLM in Table A2. For $p = 2/3$, the tests show that each reasoning LLM significantly outpaces every standard model ($p < 0.05$). For $p = 4/3$, the tests verify that GPT-o1 and Cl-4ST require significantly less learning due to their rapid convergence. However, Ge-2T converges more slowly relative to the other reasoning LLMs and Cl-3S, explaining why Ge-2T has higher average learning than those models. As Ge-1.5 appears to converge slowly and even diverge in Rounds 7 and 8, Ge-2T learns more than Ge-1.5.

²⁵To ensure our results are robust to near-zero denominators, we re-plot the learning trajectories (Appendix, Section B.5.1, Figure B6) excluding rounds with near-zero error gaps. The exclusion cutoff is defined as the 5th percentile of the non-zero absolute errors (i.e., $|w_{r-1} - g_{i,r-1}|$). As in Figure 2, we retain exact equilibrium rounds where $\alpha_{i,r} = 0$. As the differences between the filtered and original trajectories are minimal, we confirm that our findings are not driven by extreme values from small-denominator rounds.

²⁶This is intuitive as, in principle, it takes fewer levels/steps of reasoning to achieve the rational prediction when $p = 4/3$ compared to $p = 2/3$.

²⁷Recall that in pBCG, all players have the same target multiplier – 1/2, 2/3, or 4/3 – and the same range of choices, $[0, 100]$.

²⁸A game is called *dominance-solvable* if the iterated elimination of dominated strategies results in a unique Nash equilibrium, as in pBCG and GG, but not in MRG.

no matter the number of levels/steps of reasoning.^{29,30}

Game. There are two players, 1 and 2. Player 1 makes a *guess*, x^1 , within a range defined by a lower limit, a^1 , and an upper limit, b^1 . Similarly, for player 2, $x^2 \in [a^2, b^2]$. Each player i 's guess should be as close as possible to their target number, which is their target multiplier p^i multiplied by the other player j 's guess, that is, $p^i \times x^j$. Specifically, player i 's payoff is given by

$$\pi^i = \max\{0, 200 - d^i\} + \max\{0, 100 - d^i/10\},$$

where d^i is the absolute difference value between player i 's guess and player i 's target number, defined as $d^i \equiv |x^i(a^i, b^i) - p^i \cdot x^j(a^j, b^j)|$.

The payoff function π^i is uniquely maximized when $d^i = 0$ for player i and provides an additional incentive for a closer guess $d^i \leq 200$. Hence, unlike the pBCG, each player has a unique best response to every opponent's guess in the GG.

Theoretical Predictions. Predictions under the level- k model are made in a similar way to the pBCG, illustrated with an example. Suppose player 1 has a choice range in $[100, 900]$ with $p^1 = 0.5$ and player 2 has a choice range in $[100, 500]$ with $p^2 = 1.5$. An L_0 type randomizes a guess within their choice range. Hence, $l_0^1 = (100 + 900)/2$ and $l_0^2 = (100 + 500)/2$. An L_1 type believes that the opponent is an L_0 type and best responds to their belief. The best guess for player 1 with L_1 is $0.5 \times l_0^2 = 150$, which is within the range of player 1's choice. Hence, $l_1^1 = 150$. Similarly, the best guess for player 2 with L_1 is $1.5 \times l_0^1 = 750$. As this is not within player 2's choice range, l_1^2 is 500 which is the closest number to 750 within $[100, 500]$. This process continues, generating predictions for each level of reasoning in the level- k model for both players until the Nash equilibrium is reached.

Similarly, in the CH model, the reasoning step s_k for $k = 1, 2, \dots$, is defined as

$$(s_k^1, s_k^2) = \left(p^1 \cdot \sum_{j=0}^{k-1} f_k(j; \tau) s_j^2, p^2 \cdot \sum_{j=0}^{k-1} f_k(j; \tau) s_j^1 \right),$$

where $f_k(j; \tau)$ is the conditional probability for a step- k player, as defined in Equation (1). As before, each s_k^i must be restricted to the player's choice range $[a^i, b^i]$.

The complete sets of predictions from the level- k and the CH models for the different

²⁹Recall the pBCG with a choice range of $[0, 100]$ is, in principle, dominance solvable only in an infinite number of rounds for targets that are less than one ($p < 1$) meaning we may not be able to pin down precisely the exact levels/steps of reasoning required to reach equilibrium. In practice, however, subjects/LLMs most likely choose the Nash equilibrium response once their iterative reasoning has converged sufficiently close to the equilibrium value of 0.

³⁰From a practical standpoint, the GG is not as commonly studied in the literature, suggesting LLMs will not be trained on many sources involving the GG as compared to the pBCG; this is also true for the MRG, analyzed in the next section.

parameters (provided in Table A6) used in the GGs are given in Tables A7 and A8, respectively.

Estimation. As with the pBCG, we estimate the (same) parameters for both the level- k and CH models by MLE. The notable difference here is that we collect data sequentially for the 16 rounds of the GG. As we collect this sequence of responses 100 times for each LLM, we are able to estimate both models on an individual-response level. In other words, we estimate each model for every sequence of responses and for our final estimates, we take the average of the parameters by LLM.³¹ The interpretation of both models remains the same. A detailed description of the estimation for both models is provided in Section B.4.2 of Appendix B.

4.1 Design

Different parameters in (a^i, b^i, p^i) for players $i = 1, 2$ constitute different games. We follow precisely the 16 GGs used in Costa-Gomes and Crawford (2006). Each player has a target multiplier from $\{0.5, 0.7, 1.3, 1.5\}$, a lower limit from $\{100, 300\}$, and an upper limit from $\{500, 900\}$. The complete set of parameters, with the order of the games played, is provided in Table A6. For each LLM, we collect 100 sets of responses to the 16 GG tasks, using the same experimental design as Costa-Gomes and Crawford (2006), specifically, (i) no feedback is provided between tasks to suppress repeated-game effects, experience-based learning, and experimentation, (ii) LLMs retain the full history of their own prior tasks and choices reflecting behavior under accumulated context rather than strictly independent, zero-shot observations.

4.2 Results

As with the pBCG, we conducted a series of preliminary questions to test whether LLMs understand the rules of the games and whether they choose correct best responses when given the action of the other player. All LLMs answered questions correctly with respect to rules, however only the reasoning LLMs and Cl-3S consistently provided correct answers to the best-response questions, unlike the other LLMs with an approximately 100% error rate. Given this, we collect data focusing only on the reasoning LLMs and Cl-3S.

From Figure 3, the majority of responses for Cl-3S are consistent with L_0 and L_1 behavior with virtually no higher-order strategic reasoning. Only the reasoning LLMs exhibit any substantial proportion of L_∞ behavior, with far fewer L_0 and L_1 responses for GPT-o1. The level- k proportions are also estimated using data from Costa-Gomes and Crawford (2006)

³¹Naturally, another way to approach the estimation is to aggregate the data for each round, and estimate the model for each of the 16 rounds of the GG, as if each round were its own one-shot game. However, this does not make use of the sequential nature in which the GG is played in Costa-Gomes and Crawford (2006), which we attempt to stay as close to as possible, in terms of experimental design.

($N = 88$). Estimates suggest that Ge-2T performs similarly to human subjects, with Cl-3S performing worse and GPT-o1 and Cl-4ST demonstrating more strategic reasoning than human subjects, strongly in favor of GPT-o1. Table A3 reports frequencies of round-by-round comparisons with human subjects, showing that GPT-o1 and Cl-4ST demonstrate more strategic reasoning than Cl-3S and Ge-2T in 12 and 11 out of 16 rounds, respectively. The results from the CH model are in line with the level- k model. On average, GPT-o1 exhibits approximately 3 steps of reasoning, which is roughly 2 more steps of reasoning than Cl-3S and 1 more step than Ge-2T and Cl-4ST. Human subjects from Costa-Gomes and Crawford (2006) exhibit just under two steps, suggesting lower strategic reasoning than GPT-o1 but levels comparable to those of the other reasoning LLMs.

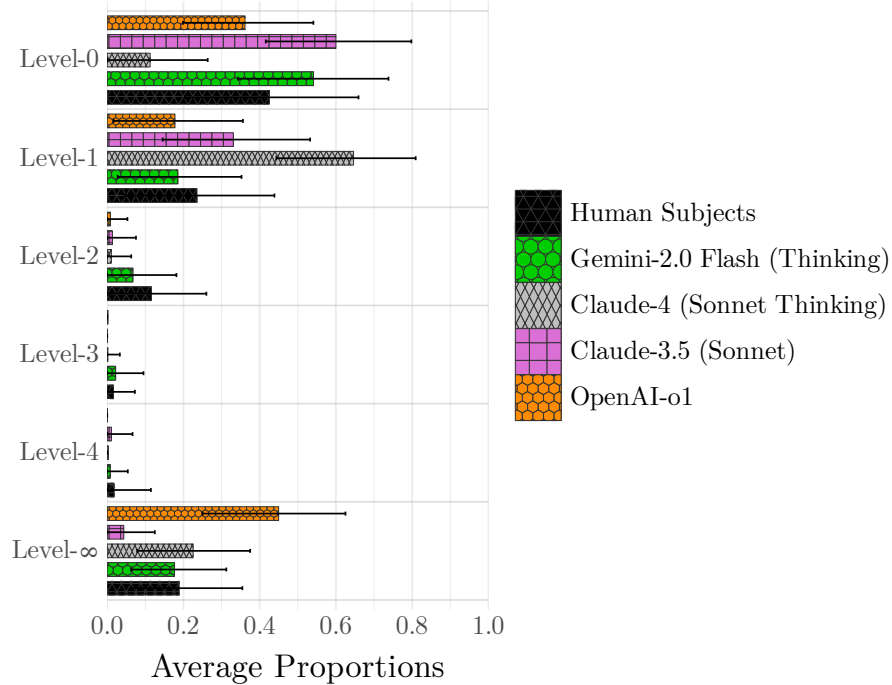


Figure 3: Estimates of the proportions of level- k types for the GG with bootstrapped 95% confidence intervals.

5 Money Request Game

The MRG offers several advantages over the pBCG and GG. From a theoretical standpoint, its simplicity triggers behavior associated with the level- k reasoning with an intuitive L_0 assumption and clearly separates reasoning levels within the action space. Additionally, there is no pure Nash equilibrium, requiring players to depend on more sophisticated strategic

	τ
GPT-o1	2.84 (1.78,4.08)
Cl-3S	0.69 (0.00,1.43)
Cl-4ST	1.94 (1.29,2.91)
Ge-2T	1.51 (0.68,2.51)
Human Subjects	1.78 (0.81,3.00)

Table 2: Estimates of τ from the CH model for the GG.

reasoning.^{32,33}

Game. Two players simultaneously choose an integer between 11 and 20. Each player is awarded an amount of money corresponding to the number they choose. Additionally, if a player selects a number exactly one less than the opponent’s choice, the player receives an extra payment of 20.

Theoretical Predictions. The choice of 20 plays a role as an anchor, which we denote by L_0 . Clearly, the choice of 20 is not driven by uniform randomization, but it is a *safe* choice since it does not involve any strategic concern in the sense that a player can secure the payoff of 20 from the choice of 20 (but without any chance of receiving the bonus payoff). As such, the choice of 19 is the best response to 20, the choice of 18 is the best response to 19,..., and the choice of 11 is the best response to 12.

From this observation, the level- k model predictions for the reasoning levels l_k are as follows.

$$l_0 = 20, \quad l_1 = 19, \quad l_2 = 18, \quad \dots, \quad l_9 = 11.$$

Similarly, in the CH model, the reasoning step s_k for $k = 1, 2, \dots$, is defined as

$$s_k = \text{round} \left(\sum_{j=0}^{k-1} f_k(j; \tau) s_j \right) - 1,$$

where $\text{round}(x)$ outputs the nearest integer to x , and $f_k(j; \tau)$ is the conditional probability

³²Choosing any particular number with certainty for each player is not a Nash equilibrium. Instead, each player would need to randomize over actions according to a pre-specified distribution.

³³We focus on Game 1 of Arad and Rubinstein (2012). Game 3, which is a variation of Game 1, is included in Section B.1 of Appendix B; the results are approximately the same across games.

for a step- k player, as defined in Equation (1).

Estimation. As with the pBCG and GG, we use MLE to estimate the level- k model and the CH model, analogously to what was done previously. We keep the same number of levels/steps as the pBCG but replace L_∞ with L_4 , as there is no pure Nash equilibrium. We still accommodate behavior which is not predicted by the level- k or CH model by including a purely random type. However, as L_0 behavior can be very distinct from purely random behavior, we specifically distinguish between these two types.³⁴ Given the finite nature of the actions and theoretical predictions in the MRG, MLE is more appropriately specified using the responses and best guesses directly, as opposed to a noisy best guess. A detailed description of the estimation for both models is provided in Section B.4.3 of Appendix B.

5.1 Results

Each LLM consistently provided correct answers, both when asked about the game rules and when responding to questions about best responses.

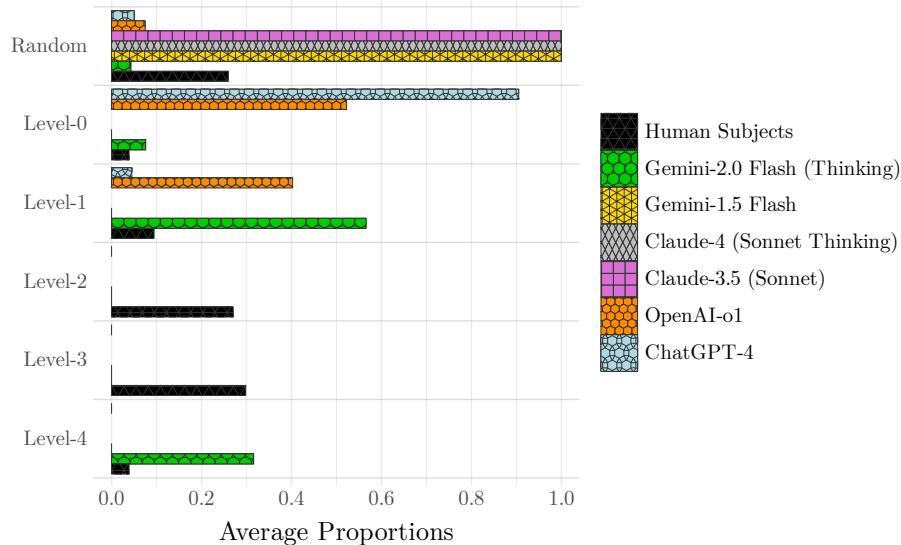


Figure 4: Estimates of the proportions of level- k types for MRG.

From Figure 4, the distinction between standard LLMs and reasoning LLMs is not as clearly dichotomous. While it is clear that Ge-2T exhibits the highest strategic reasoning (and overall still weakly in favor of reasoning LLMs), no other LLM shows any substantial strategic reasoning beyond L_1 with both GPT-o1 and Cl-4ST exhibiting substantially high

³⁴ Another reason for doing this is to allow 15 to be treated as a choice consistent with the random choice, and lower choices, to be considered as random behavior, rather than a very high stage of reasoning. This is consistent with the empirical results in Arad and Rubinstein (2012) with a minority of subjects choosing lower than 15.

	MRG
GPT-4	0.91 (0.86,0.97)
GPT-o1	1.21 (1.11,1.31)
Cl-3S	0.00 (0.00,0.00)
Cl-4ST	0.00 (0.00,0.70)
Ge-1.5	0.00 (0.00,0.00)
Ge-2T	1.00 (1.00,1.22)
Human Subjects	3.14 (2.17,3.65)

Table 3: Estimates of τ from the CH model for MRG.

levels of random/ L_0 behavior. Using data from Arad and Rubinstein (2012) as the human-subject benchmark, we find that human subjects are mainly L_2 and L_3 , suggesting all LLMs are vastly outperformed except Ge-2T which exhibits comparable or higher levels of reasoning than human subjects.³⁵

From the estimates of τ in Table 3, both GPT-o1 and Ge-2T exhibit the highest strategic reasoning followed by GPT-4, with the remaining models showing no steps of reasoning. These estimates are largely in agreement with the level- k model. The largest discrepancy is for Ge-2T, for which the level- k estimates show substantially higher-level behavior while the CH estimate is only $\tau = 1$. This discrepancy illustrates the CH model’s sensitivity to multimodal response patterns. When responses contain both substantial lower-step behavior and isolated higher-step behavior, the Poisson restriction may lead to a relatively low best-fitting τ , even if the level- k estimates reveal nontrivial higher-level responses. Thus, the low estimate of τ should be interpreted as the best-fitting CH summary of this response distribution, rather than a direct measure of LLMs’ strategic capacity. Compared to human subjects, LLMs are clearly outperformed under the CH estimates, with human subjects exhibiting just over 3 steps of reasoning.

³⁵Using data from Arad and Rubinstein (2012) ($N = 108$), we also conduct similar one-sided KS tests as with the pBCG. Excluding LLMs that exhibited random behavior by our definition (Ge-1.5) and responses from both Cl-3S and Cl-4ST that are random (by not exhibiting L_0 to L_4 behavior), all p-values were less than 0.005. This implies that human subject responses were cumulatively lower than LLMs, thus suggesting LLMs were outperformed across the whole distribution of responses.

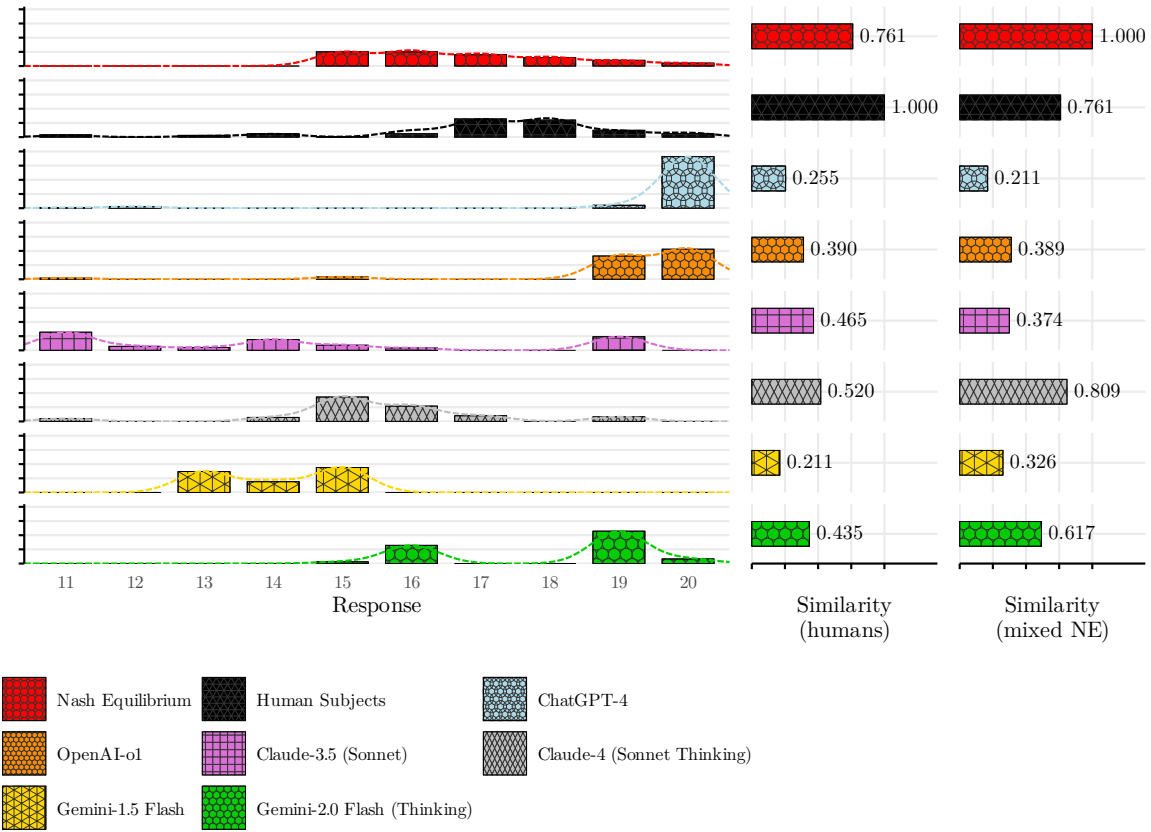


Figure 5: Histograms of responses in the MRG (left panel) alongside similarity measures (middle and right panels). The horizontal bars reflect the magnitude of similarity between the LLMs’ response distributions and those of human subjects and the mixed Nash equilibrium, respectively, measured as 1 minus the Jensen-Shannon divergence. Dashed continuous density plots are overlaid on the histograms.

To check whether LLMs (alongside human subjects) are, on aggregate, playing the mixed Nash equilibrium, we plot the distribution of their responses in Figure 5. First, in tandem with the previous analysis, when we compare all response distributions with the human-subject response distribution, the mixed Nash equilibrium distribution is the most similar to the human subjects’ response distribution; this is to say that aggregate LLM responses are less similar to human responses than human responses are to the mixed Nash equilibrium distribution.³⁶ Likewise, when we compare all response distributions with the mixed

³⁶Similarity, as reported in Figure 5, is measured using the Jensen-Shannon divergence (JSD) which is a symmetric and smoothed adaptation of the Kullback-Leibler divergence (KLD), used to quantify how much two probability distributions differ from one another. Specifically, JSD is constructed by averaging the KLD of each distribution, relative to their mixture distribution. A key advantage of this construction is that when the underlying KLD is computed using base-2 logarithms, the resulting JSD is strictly bounded between 0 and 1. As the JSD measures divergence, we define similarity as $1 - \text{JSD}$.

Nash equilibrium distribution, except Cl-4ST, human subjects are still more similar to the mixed Nash equilibrium distribution than the other LLMs are to the mixed Nash equilibrium. Despite Cl-4ST responses exhibiting high similarity with the mixed Nash equilibrium distribution, permutation tests reject the null hypothesis that its responses are drawn from the same underlying distribution (Table A4). Additionally, two-sided KS tests against the mixed Nash equilibrium likewise all reject the null hypothesis of identical distributions at any reasonable significance level (Table A5).

6 Discussion

In this paper, we evaluate the strategic reasoning capabilities of standard and reasoning LLMs across three well-known games in behavioral economics. Reasoning LLMs generally exhibit greater strategic sophistication than standard LLMs, but the intensity of this difference varies across games. In the pBCG, reasoning LLMs exhibited higher levels of iterated reasoning in the one-shot setting and converge more quickly toward the rational prediction in the repeated setting with feedback. In the GG, the distinction was especially pronounced, as reasoning LLMs exhibited substantially higher-order reasoning, while most standard LLMs struggled with best-response calculations. However, in the MRG, the difference between reasoning and standard LLMs was smaller. Thus, we conclude that reasoning LLMs demonstrate stronger strategic reasoning than standard LLMs, but that this advantage is sensitive to the strategic environment.

The difference between reasoning and standard LLMs is not explained by rule comprehension alone. In the pBCG, all LLMs understand the rules and calculate appropriate best responses when opponents’ actions are specified. Yet, standard LLMs perform substantially worse when they have to anticipate others’ reasoning. Therefore, reasoning LLMs appear better able to use analytical calculation for recursive strategic reasoning.

The comparison with human subjects also shows that LLM strategic reasoning is task-dependent rather than uniformly superior or inferior to human reasoning. Reasoning LLMs often matched or exceeded human benchmarks in the pBCG and GG, where a higher level of rationality is closely associated with systematic iteration toward a well-defined prediction. In contrast, human subjects clearly outperformed LLMs in the MRG under the CH estimates, although Ge-2T appeared comparable under the level- k analysis. This contrast suggests that reasoning LLMs perform relatively well in structured environments that reward iterative reasoning, while human subjects perform better when strategic behavior is more heterogeneous and less predictable.

The results of the repeated pBCG provide a complementary perspective. Even standard LLMs adjusted their responses toward rational predictions when feedback is available, while reasoning LLMs generally converged more rapidly. This pattern is consistent with evidence

that repeated interaction and feedback affect strategic adjustment among human subjects (Ho, Camerer and Weigelt, 1998; Weber, 2003). It also suggests that LLM strategic performance may be more reliable in repeated settings with informative feedback than in one-shot settings where strategic errors are difficult to correct.

At the same time, the $p = 4/3$ condition in the pBCG highlights a limitation in the robustness of LLMs’ strategic reasoning. In this treatment, all LLMs struggled when the direction of iterative reasoning differed from the more conventional downward-reasoning version of the pBCG (such as the $p = 2/3$ or $p = 1/2$ conditions). This is consistent with recent evidence that LLM performance can deteriorate under controlled modifications of otherwise familiar reasoning problems (Shojaee et al., 2026; Mirzadeh et al., 2025). Our results suggest a similar finding in strategic environments. Further, although reasoning LLMs generally exhibit greater strategic sophistication than standard LLMs, this strength does not necessarily transfer easily. In particular, the lower strategic reasoning in the less conventional $p = 4/3$ pBCG condition suggests that LLM performance may depend partly on familiar task structures, such as the *downward* reasoning required in pBCGs with $p = 2/3$, rather than reflecting fully generalizable strategic reasoning. One possible explanation is that LLMs may have had greater prior exposure to the $p = 2/3$ condition, which is much more common in the literature, than to the $p = 4/3$ condition.

A related caveat concerns possible benchmark contamination. Human subjects in the benchmark participated in these games in controlled experimental settings, whereas LLMs are very likely to have been exposed (during pre-training) to prior analyses, and/or solution patterns from those games. Naturally, as the training data of the LLMs examined here are unavailable, we cannot rule out such prior exposure. Accordingly, comparisons with human subjects should be interpreted as behavioral benchmarks rather than as clean zero-shot comparisons under identical informational conditions.

These findings are especially relevant for potential real-world uses of LLM strategic reasoning. A growing literature considers LLM-based agents in settings where anticipating the actions of others is important (Duan et al., 2024a; Lorè and Heydari, 2024; Duan et al., 2024b; Bousetouane, 2025; Singh et al., 2024). In negotiation, supply-chain coordination, or financial decision support, LLMs could assist decision-makers by anticipating reactions to alternative actions and comparing possible strategic responses. Strategic reasoning may also matter in high-stakes defense decision support, where AI-assisted systems may help human operators assess evolving threats and possible adversarial responses under severe time constraints. In such settings, the relevant capability is not simply predicting outcomes, but reasoning about other agents under uncertainty. Our findings therefore suggest that LLM strategic capabilities should be validated across diverse environments and used with appropriate human oversight.

Finally, our comparisons with humans serve as benchmarks of strategic sophistication

rather than tests of whether LLMs reproduce human behavior. Gao et al. (2025) argue that human-like behavior in selected tasks is insufficient for treating LLMs as human surrogates. Our results address a distinct but related question by showing that LLMs’ performance relative to humans varies across strategic environments. Whether LLMs can serve as human surrogates therefore remains an open question for future research.

Data Availability

All datasets used are publicly available *here*. This includes all estimates and figures contained in this article.

Code Availability

Our code is publicly available *here*.

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Additional Information

Supplementary tables are provided in Appendix A.

Supplementary information is available for this paper in Appendix B.

A Supplementary Tables

LLM	Game (pBCG)	H0: Two-sided	H0: Not Less	H0: Not Greater	More rational
GPT-4	Median	0	0	0.243	Human
GPT-4	n=unspecified	0	0	0.57	Human
GPT-4	n=2	0	0	0	-
GPT-4	p=1/2	0	0	0.009	-
GPT-4	p=2/3	0	0.192	0	LLM
GPT-4	p=4/3	0	0.361	0	Human
GPT-o1	Median	0	1	0	LLM
GPT-o1	n=unspecified	0	1	0	LLM
GPT-o1	n=2	0	1	0	LLM
GPT-o1	p=1/2	0.011	0.005	0.048	-
GPT-o1	p=2/3	0	0.123	0	LLM
GPT-o1	p=4/3	0	1	0	Human
Cl-3S	Median	0.001	0.001	0.007	-
Cl-3S	n=unspecified	0	0	0.010	-
Cl-3S	n=2	0	0	0	-
Cl-3S	p=1/2	0	0	0	-
Cl-3S	p=2/3	0	0.028	0	-
Cl-3S	p=4/3	0	0.285	0	Human
Cl-4ST	Median	0	0	0.003	-
Cl-4ST	n=unspecified	0	0	0	-
Cl-4ST	n=2	0	0.001	0	-
Cl-4ST	p=1/2	0	0.762	0	LLM
Cl-4ST	p=2/3	0	1	0	LLM
Cl-4ST	p=4/3	0.053	0.027	0.474	LLM
Ge-1.5	Median	0	0	0.2	Human
Ge-1.5	n=unspecified	0	0	0.153	Human
Ge-1.5	n=2	0	0	0	-
Ge-1.5	p=1/2	0	0	0.555	Human
Ge-1.5	p=2/3	0	0	0	-
Ge-1.5	p=4/3	0	0.113	0	Human
Ge-2T	Median	0	0	0	-
Ge-2T	n=unspecified	0	0	0	-
Ge-2T	n=2	0	0	0	-
Ge-2T	p=1/2	0	0.078	0	LLM
Ge-2T	p=2/3	0	0.26	0	LLM
Ge-2T	p=4/3	0	0.522	0	Human

Table A1: KS tests for raw responses from the pBCG. The distribution of responses from human subjects is more rational if it unambiguously first-order stochastically dominates according to the one-sided KS tests.

		$p = 2/3$			$p = 4/3$		
RLLMs	SLLMs	$\bar{\alpha}_{RLLMs}$	$\bar{\alpha}_{SLLMs}$	H0: Not Greater	$\bar{\alpha}_{RLLMs}$	$\bar{\alpha}_{SLLMs}$	H0: Not Greater
GPT-o1	GPT-4	1.57	0.68	0	0.10	0.44	1
GPT-o1	Cl-3S	1.57	0.63	0	0.10	0.35	1
GPT-o1	Ge-1.5	1.57	0.25	0	0.10	0.23	1
Cl-4ST	GPT-4	1.03	0.68	0	0.20	0.44	1
Cl-4ST	Cl-3S	1.03	0.63	0.022	0.20	0.35	1
Cl-4ST	Ge-1.5	1.03	0.25	0	0.20	0.23	0.925
Ge-2T	GPT-4	1.23	0.68	0	0.44	0.44	0.625
Ge-2T	Cl-3S	1.23	0.63	0.003	0.44	0.35	0
Ge-2T	Ge-1.5	1.23	0.25	0	0.44	0.23	0

Table A2: Pairwise one-sided Wilcoxon rank-sum tests comparing the average learning coefficient ($\bar{\alpha}$) of reasoning LLMs (RLLMs) against standard LLMs (SLLMs). The hypothesis tests whether the reasoning model exhibits a significantly greater learning rate.

LLM	LLMs more rational	Rounds
GPT-o1	12	1-11, 14
Cl-3S	3	2, 5, 9
Cl-4ST	11	1-2, 4, 6-10, 13-14, 16
Ge-2T	4	1-3, 8

Table A3: Frequency of LLMs providing more Nash equilibrium responses than human subjects (out of 16), and in which specific rounds. LLMs are considered more rational if the number of Nash equilibrium responses is strictly larger than that of human subjects.

	LLM	Similarity	p -value	H0: Same Distribution
Human	GPT-4	0.255	0	Reject
	GPT-o1	0.390	0	Reject
	Cl-3S	0.465	0	Reject
	Cl-4ST	0.520	0	Reject
	Ge-1.5	0.211	0	Reject
	Ge-2T	0.435	0	Reject
Mixed NE	GPT-4	0.211	0	Reject
	GPT-o1	0.389	0	Reject
	Cl-3S	0.374	0	Reject
	Cl-4ST	0.809	0	Reject
	Ge-1.5	0.326	0	Reject
	Ge-2T	0.617	0	Reject

Table A4: Permutation tests (100,000 iterations) for comparing the response distributions of each LLM in the MRG against human subjects and the mixed Nash equilibrium (NE), respectively.

	LLM	H0: Two-sided	H0: Not Less	H0: Not Greater	More rational
Human	GPT-4	0	0	0.870	Human
	GPT-o1	0	0	1	Human
	Cl-3S	0	0	0.590	-
	Cl-4ST	0	0.833	0	-
	Ge-1.5	0	0.644	0	-
	Ge-2T	0	0	0.026	-
Mixed NE	GPT-4	0	-	-	-
	GPT-o1	0	-	-	-
	Cl-3S	0	-	-	-
	Cl-4ST	0	-	-	-
	Ge-1.5	0	-	-	-
	Ge-2T	0	-	-	-

Table A5: KS tests for raw responses from the MRG. The human baseline distribution is considered more rational if it unambiguously first-order stochastically dominates the LLM response distribution based on the one-sided KS tests (excluding Ge-1.5, Cl-3S, and Cl-4ST).

	Player 1			Player 2		
Games	Targets	Lower Limits	Upper Limits	Targets	Lower Limits	Upper Limits
1	1.3	300	900	1.5	300	500
2	0.7	100	500	1.5	100	500
3	1.5	100	900	0.7	300	500
4	1.5	100	500	0.5	100	900
5	0.5	100	900	1.5	100	500
6	0.5	100	900	0.7	100	500
7	1.3	100	900	0.7	300	900
8	0.5	100	900	0.7	300	500
9	1.5	100	500	0.7	100	500
10	0.7	300	900	1.3	100	900
11	0.7	300	500	0.5	100	900
12	0.7	300	500	1.5	100	900
13	1.3	300	900	1.3	300	900
14	0.7	100	500	0.5	100	900
15	1.3	300	900	1.3	300	900
16	1.5	300	500	1.3	300	900

Table A6: Task parameters from Costa-Gomes and Crawford (2006) used for the GG.

Games	l_0	l_1	l_2	l_3	l_4	l_5	l_6	l_7	l_8	l_9	l_{10}	l_{11}
1	600	520	650*									
2	300	210	315	220.5	330.75	231.53	347.29	243.10	350*			
3	500	600	525	630	551.25	661.50	578.81	694.58	607.75	729.30	638.14	750*
4	300	500	225	375	168.75	281.25	150*					
5	500	150	250	112.50	187.50	100*						
6	500	150	175	100*								
7	500	780	455	709.80	414.05	645.92	390*					
8	500	200	175	150*								
9	300	450	315	472.5	330.75	496.13	347.29	500*				
10	600	350	546	318.50	496.86	300*						
11	400	350	300*									
12	400	350	420	367.50	441	385.88	463.05	405.17	486.20	425.43	500*	
13	600	780	900*									
14	300	350	105	122.50	100*							
15	600	780	900*									
16	400	500*										

Table A7: Predictions from the level- k model for the 16 GGs. Asterisk refers to reaching the Nash equilibrium action.

Games	s_0	s_1	s_2	s_3	s_4	s_5
1	600	520	598	614.14	618.96	620.44
2	300	210	273	268.44	268.26	268.30
3	500	600	555	565.24	568.08	569.00
4	300	500	435	397.76	387.11	383.69
5	500	150	210	212.33	210.52	209.71
6	500	150	165	145.29	137.59	134.97
7	500	780	585	592.10	591.07	590.46
8	500	200	185	174.14	170.90	169.89
9	300	450	369	381.57	384.40	385.26
10	600	350	467.60	449.57	444.85	443.36
11	400	350	300*			
12	400	350	392	390.91	391.55	391.85
13	600	780	900*			
14	300	350	203	175.84	165.89	162.55
15	600	780	900*			
16	400	500*				

Table A8: Example predictions from the CH model for the 16 GGs at $\tau = 1.5$.

B Supplementary Methods and Notes

B.1 Supplementary MRG

Here, we analyze Game 3 of Arad and Rubinstein (2012), which is a costless iteration version of the MRG in the main text. Game 1 (main text) and Game 2 of Arad and Rubinstein (2012) are game-theoretically identical. The only difference between Games 1 and 2 is the increased salience of L_0 behavior.³⁷ The payoff structure of Game 3 is a slight variation from Game 1/2, with a different equilibrium prediction, but is ultimately similar to MRG.

MRG (Game 3). Two players simultaneously choose an integer between 11 and 20. A player is awarded 20 if they choose 20. Choosing any other number results in receiving 17. Additionally, if a player selects a number exactly one less than the opponent’s choice, the player receives an extra payment of 20.

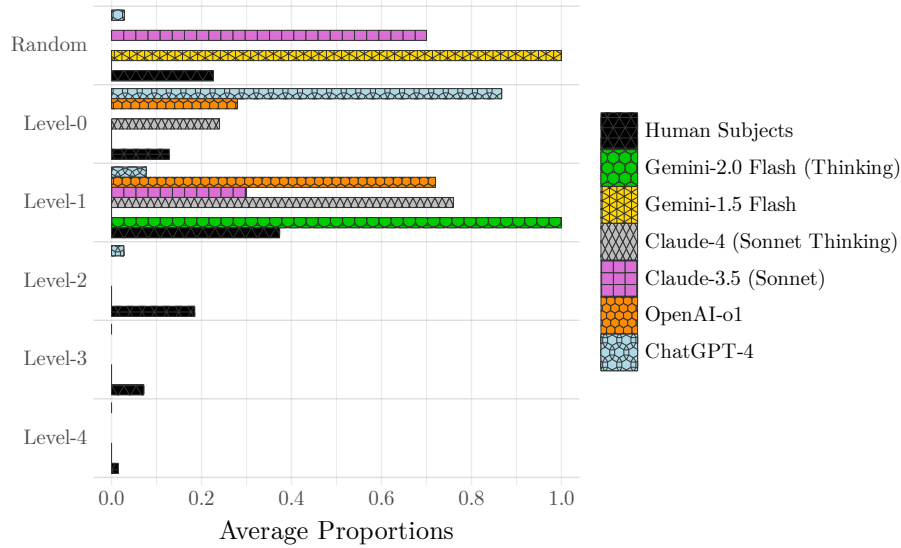


Figure B1: Estimates of the proportions of level- k types for MRG (Game 3).

For the MRG (Game 3), the level- k results are largely the same for GPT-4 and Ge-1.5 as in the MRG. Ge-2T is less sophisticated, with the majority of answers being consistent with L_1 behavior, with a large reduction in higher-step responses (Figure B1). For GPT-o1, Cl-3S, and Cl-4ST we observe increases in strategic reasoning that are in line with the original formulation of MRG (Game 3). This is because undercutting as a strategy is no longer costly. Unlike in MRG when players may balance between choosing a higher number

³⁷ Additionally, some LLMs answered the understanding questions poorly for Game 1 when asked about best responses to 11 (the lowest number). To remedy this, we make the instructions clearer by being specific about the payoff when 11(/20) is chosen and 11(/20) is played by the other participant. As such, adding this to Game 1’s instructions essentially replicates Game 2’s instructions.

	MRG (Game 3)
GPT-4	1.00 (0.93,1.06)
GPT-o1	1.60 (1.53,1.67)
Cl-3S	0.00 (0.00,0.00)
Cl-4ST	1.64 (1.57,1.70)
Ge-1.5	0.00 (0.00,0.00)
Ge-2T	1.89 (1.89,1.89)

Table A9: Estimates of τ from the CH model for MRG (Game 3).

(between 11 and 19) yielding a higher payoff, and undercutting to get the bonus payoff, MRG (Game 3) removes the marginal incentive in choosing a higher number. This suggests that we should observe increases in strategic reasoning in MRG (Game 3) relative to MRG if there is sufficient understanding of the changed payoff structure, which we only observe for GPT-o1, Cl-3S, and Cl-4ST.³⁸ When compared with Arad and Rubinstein (2012) ($N = 53$), human subjects are mainly random, L_1 and L_2 , suggesting LLMs are all still vastly outperformed by human subjects.^{39,40}

The estimates of τ are in line with the level- k explanations above (Table A9).⁴¹ Cl-4ST and Ge-2T have apparent jumps in steps of reasoning owing to the fact that there is now a clear peak in the distribution of responses around the step-1/2 responses, allowing a much better fit of the CH model, given the Poisson distribution assumption.⁴² Permutation tests of similarity follow the same conclusions as the MRG (Game 1) as in the main text.

³⁸Confirmed by one-sided KS tests at the 5% significance level.

³⁹The estimated level- k model proportions (random, L_0 to L_4) are 0.23, 0.13, 0.37, 0.18, 0.07, 0.02, respectively.

⁴⁰Conducting similar one-sided KS tests using data from Arad and Rubinstein (2012), again excluding Ge-1.5, Ge-2T and Cl-3S, shows that human subjects outperformed LLMs (Table A10).

⁴¹The CH model estimate from human subjects is $\tau = 2.45$; $[1.16, 2.45]_{95\% \text{ CI}}$.

⁴²A similar explanation for why Cl-3S does not exhibit higher steps of reasoning applies, despite showing an increase in L_1 responses. As the majority of the answers are consistent with random behavior, with no responses in between random and L_1 behavior, the CH model fits better for a low τ .

LLM	H0: Two-sided	H0: Not Less	H0: Not Greater	More rational
GPT-4	0	0	1	Human
GPT-o1	0	0	1	Human
Cl-3S	0.024	0.016	0.069	-
Cl-4ST	0	0	1	Human
Ge-1.5	0	0.619	0	-
Ge-2T	0	0	0.009	-

Table A10: KS tests for raw responses from the MRG (Game 3). The distribution of responses from human subjects is more rational than Ge-1.5, Ge-2T, and Cl-3S, if it is unambiguously first-order stochastically dominated from the one-sided KS tests.

B.2 Understanding Questions

The following lists all of the understanding questions asked to each LLM related to best responses (BR) and the rules of each game. The system prompt is the same for all questions (as written in the paper). Each question is repeatedly asked in isolation (no history of previous answers).

B.2.1 pBCG

BR for 2-person ($p=2/3$)

Including yourself, there are 2 participants in total. The other participant you will be playing with is a human decision maker. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$ multiplied the average of all the numbers provided by the participants (including your own). For example, if you choose 10 and your opponent chooses 20, then you will be the winner, as 10 is closer to $15 \cdot 2/3 = 10$ (as 15 is the average and $2/3$ is the target). Or, if you choose 30 and your opponent chooses 20, your opponent will be the winner, as 20 is closer to $25 \cdot 2/3 = 16.67$ (as 25 is the average and $2/3$ is the target). In case of multiple winners, only one participant will be randomly chosen to win. Suppose your opponent chooses 20. What number from 0 to 100 earns you the most?

Best responses: $\{0, 1, \dots, 19\}$

BR for 11-person ($p=1/2$)

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to

$1/2$ multiplied by the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 10 will be the winner, as 10 is closer to $20 \cdot 1/2 = 10$ (as 20 is the average and $1/2$ is the target). Or, if you choose 25 and your opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then you will be the winner, as 25 is closer to $50 \cdot 1/2 = 25$ (as 50 is the average and $1/2$ is the target). In case of multiple winners, only one participant will be randomly chosen to win. Suppose your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50. What number from 0 to 100 earns you the most?

Best responses: {19,20}

BR for Baseline: 11-person ($p=2/3$)

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$ multiplied by the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then you will be the winner, as 14 is closer to $20 \cdot 2/3 = 13.33$ (as 20 is the average and $2/3$ is the target). Or, if you choose 25 and your opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then the opponent with 40 will be the winner, as 40 is closer to $50 \cdot 2/3 = 33.3$ (as 50 is the average and $2/3$ is the target). In case of multiple winners, only one participant will be randomly chosen to win. Suppose your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50. What number from 0 to 100 earns you the most?

Best responses: {22,23,...,31}

BR for 11-person ($p=4/3$)

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $4/3$ multiplied by the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 30 will be the winner, as 30 is closer to $20 \cdot 4/3 = 26.67$ (as 20 is the average and $4/3$ is the target). Or, if you choose 25 and your

opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then the opponent with 70 will be the winner, as 70 is closer to $50 \cdot \frac{4}{3}$ (as 50 is the average and $\frac{4}{3}$ is the target). In case of multiple winners, only one participant will be randomly chosen to win. Suppose your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50. What number from 0 to 100 earns you the most?

Best responses: $\{51, 52, \dots, 62\}$

BR for n-person ($p=\frac{2}{3}$)

Including yourself, there is a finite but unknown number n of participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $\frac{2}{3}$ multiplied by the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then you will be the winner, as 14 is closer to $20 \cdot \frac{2}{3} = 13.33$ (as 20 is the average and $\frac{2}{3}$ is the target). Or, if you choose 25 and your opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then the opponent with 40 will be the winner, as 40 is closer to $50 \cdot \frac{2}{3} = 33.3$ (as 50 is the average and $\frac{2}{3}$ is the target). In case of multiple winners, only one participant will be randomly chosen to win. In the case with $n = 11$, if your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50, what number from 0 to 100 earns you the most?

Best responses: $\{22, 23, \dots, 31\}$

BR for Median

Including yourself, there are 11 participants in total. The other participant you will be playing with is a human decision maker. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $\frac{2}{3}$ multiplied by the median of all the numbers provided by the participants (including your own). For example, if you choose 80 and your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50, then one of the opponents with 50 will be the winner, as 50 is closer to $43(\text{median}) \cdot \frac{2}{3}$. Or, if you choose 100 and your opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then the opponent with 40 will be the winner, as 40 is closer to $60(\text{median}) \cdot \frac{2}{3}$. In case of multiple winners, only one participant will be randomly chosen to win. Suppose your opponents choose 0, 80, 43, 70, 21, 33, 37, 18, 50, 50. What number from 0 to 100 earns you the most?

Best responses: $\{21, 23, \dots, 28\}$

For each of the conditions above, LLMs were also asked the following questions (repeatedly, and in isolation) related to the rules and the win conditions of the pBCG:

Rules for pBCG

1. What numbers can you choose between?
2. If the number you choose is closest to $\{p\}$'s of the $\{\text{average/median}\}$, do you win the game?
3. How many participants are there, including yourself?
4. Suppose there are 3 participants, including you, and you have to choose the number closest to $\{p\}$ of the average. The other 2 participants chose 90 and 60. Suppose you chose 30. Therefore $\{p\}$'s of the $\{\text{average/median}\}$ is $\{\{p\} \times 60\}$. Do you win or lose?^a
5. Will decisions made in previous rounds have any effect on the current round? (repeated pBCG)

^aThe question uses actual values for $\{p\}$ and $\{\text{average/median}\}$ depending on the condition. $\{p\} \times 60$ is evaluated, i.e., the actual calculation is shown.

B.2.2 GG

BR for GG

This game concerns a decision situation in which you and another person we call “s/he” (which will refer to a new person each round) separately and independently make decisions called GUESSES. Together, your and her/his guesses determine the numbers of POINTS that you and s/he earn in a round, which may be different. To choose your guesses, it may help you to understand how your and her/his guesses will determine the numbers of points that you and s/he earn in the decision situations. In each decision situation, each person has her/his own TARGET, LOWER LIMIT and UPPER LIMIT. These targets and limits may be different for you and her/him, and they may change from round to round. Otherwise, the decision situations are identical in all 16 rounds. Your and her/his targets, lower limits, and upper limits will be known to you and her/him every round. Both you and s/he will receive the same instructions and have the same information about the decision situations and the same access to your and her/his targets and limits. You (respectively, s/he) can choose your (her/his) guesses only within your (her/his) given limits for each round as explained below. After submitting guesses, you earn whichever is larger, either 0 points or 200 points minus the distance between YOUR guess and the product of YOUR

target times HER/HIS guess, PLUS whichever is larger, either 0 points or 100 points minus one-tenth (1/10th) the distance between YOUR guess and the product of YOUR target times HER/HIS guess. S/he earns whichever is larger, either 0 points or 200 points minus the distance between (1)HER/HIS guess and (2)the product of HER/HIS target times YOUR guess, PLUS whichever is larger, either 0 points or 100 points minus one-tenth (1/10th) the distance between (3)HER/HIS guess] and (4)the product of HER/HIS target times YOUR guess. That is,

$$\text{Points} = \max\{0, 200 - \text{distance}\} + \max\{0, 100 - 0.1 * \text{distance}\},$$

where distance is the absolute difference between Your guess and your Target*(His/Her Guess)

This way of determining the number of points that you and s/he earn makes the number you earn larger, the closer your guess is to your target times her/his guess; and it makes the number s/he earns larger, the closer her/his guess is to her/his target times your guess. Only the distance matters, NOT whether the difference is positive or negative. You earn the same number of points when your guess is too high by a given amount as when it is too low by the same amount. It is important to understand how your (respectively, her/his) original guesses should be chosen to stay within your (her/his) limits. This will be done as follows. If your original guess is below your lower limit, then your guess should be adjusted UP to your LOWER limit; and if your original guess is above your upper limit, then your guess should be adjusted DOWN to your UPPER limit. If, for example, your lower limit is 400 and you original guess 300, then your adjusted guess is 400. If your upper limit is original 600 and your guess 900, then your adjusted guess is 600. Her/his guesses are adjusted up or down to her/his lower or upper limits in the same way, except that her/his limits may be different. Remember you and s/he can choose any number within your and her/his limits, respectively.

Suppose that:

Your Limits & Target are: Lower Limit = 200, Upper Limit = 600, Target = 1.2

Her/His Limits & Target are: Lower Limit = 400, Upper Limit = 800, Target = 0.8

1. If s/he guesses 500, which of your guesses earns you the most points? How many points would you earn by entering that guess?
2. If you guess 400, which of her/his guesses earns her/him the most points? How many points would s/he earn by entering that guess?
3. If s/he guesses 800, which of your guesses earns you the most points?
4. If your guess is 600, which of her/his guesses earns her/him the most points?

The following are the correct responses:

1. Answer = 600, Points = 300;
2. Answer = 400, Points = 212;
3. Answer = 600;
4. Answer = 480.

Each of these 4 questions are repeatedly asked in isolation with the same set of instructions above (no history of previous answers for any of the questions).

LLMs were also asked the following questions (repeatedly, and in isolation) related to the rules and the win conditions of the pBCG:

Rules for GG

1. What numbers can you choose between?
2. Do you earn more/fewer/the same points when your guess is too high or too low by the same amount?
3. Will the targets, lower limits, and upper limits for each player be known to all players?
4. How many participants are there, including yourself?
5. Will decisions made in previous rounds have any effect on the current round?

B.2.3 MRG

BR for MRG

You are randomly matched to play a game against a human decision maker. In the game, each of you chooses an integer between 11 and 20. Each participant will receive points in the amount they have chosen. A participant will receive an additional 20 points if they choose a number exactly one less chosen by the other player. For example, if the other participant chooses X , you will get $(X-1)+20$ points if you choose $X-1$ points. For example, if your opponent chooses 14, you will get 33 points if you choose 13. Or, if the other participant chooses 11, you will get 20 points if you choose 20.

1. Suppose your opponent chooses 15. What number from 11 to 20 earns you the most?
2. Suppose your opponent chooses 11. What number from 11 to 20 earns you the most?

The following are the correct responses:

1. Answer = 14;
2. Answer = 20.

BR for MRG (Game 3)

You are randomly matched to play a game against a human decision maker. In the game, each of you chooses an integer between 11 and 20. If a participant chooses the number 20, they will receive 20 points (regardless of the other player's choice). Otherwise, if a participant chooses any other number in this range, they will receive three points less than in the case where they choose 20. However, they will receive an additional amount of 20 points if they choose a number that is lower by exactly one than the number chosen by the other player. For example, if the other participant chooses X , you will get $17+20$ points if you choose $X-1$. For example, if your opponent chooses 14, you will get $17+20$ points if you choose 13. Or, if the other participant chooses 11 points, you will get 20 points if you choose 20.

1. Suppose your opponent chooses 15. What number from 11 to 20 earns you the most?
2. Suppose your opponent chooses 11. What number from 11 to 20 earns you the most?

The following are the correct responses:

1. Answer = 14;
2. Answer = 20.

Every question, for both games, is repeatedly asked in isolation with the same set of instructions above (no history of previous answers, for any of the questions).

For each of the games above, LLMs were also asked the following questions (repeatedly, and in isolation) related to the rules and the win conditions of each game:

Rules for MRG

1. What numbers can you choose between?
2. How many participants are there, including yourself?
3. Suppose you choose 19 and the participant you have been matched with chooses 20. How many points will you receive?

B.3 Prompts

To collect data, we use the respective application programming interfaces (API) with Python/R from OpenAI, Anthropic, and Google. The exact model names of each LLM used are: gpt-4, o1-preview-2024-09-12, claude-3-5-sonnet-20241022, claude-sonnet-4-20250514, gemini-1.5-flash, gemini-2.0-flash-thinking-exp-1219. Each input prompt includes the specification of a combination of three roles: a system role, an assistant, or a user. The system role is used to guide the behavior of the LLM by providing high-level instructions and/or context. The

assistant refers to the LLM itself which generates responses for each input prompt. The user role represents the queries or prompts that are used to interact with the LLM (i.e., with the assistant). By repeated specifications of the assistant and user role within each input prompt, we are able to, in effect, have a contextual conversation. For our purpose, each input prompt will include a system role, which will be used to specify global instructions and context, and a user role, which will specify the specific rules and the request for an answer for the pBCG.⁴³

Once we knew LLMs had as good an understanding of the games as was naturally and feasibly possible, given the above questions (particularly the best response questions), we kept the wording of the prompts as close to the understanding questions. For completion, we include all prompts below.

B.3.1 pBCG

As this is a completely standard game in the behavioral economics, instructions are based closely on instructions from the literature.⁴⁴ We avoid monetary payoffs as rewards to avoid issues with incentives. We also provide examples in the instructions as LLMs had better answers to the best responses understand questions (as this helps LLMs understand what kind of calculations are involved).

B.3.2 1-shot pBCG

For the 1-shot pBCGs, the system prompt is

System

I want you to act as a human decision maker, You will be participating in 1 round of a game. You should use your best judgment to come up with responses that you like most. Always write your final answer in [].

This is identical for all 1-shot pBCG.

The following prompts include all the details of the pBCG for each condition, specified by the user role.

User for 2-person (p=2/3)

Including yourself, there are 2 participants in total. The other participant you will be playing with is a human decision maker. All participants will be asked to pick a number

⁴³Once interacted with the API using an input prompt, the answer provided by the LLM is always specified by the assistant role.

⁴⁴See Nagel (1995b); Ho, Camerer and Weigelt (1998); Bosch-Domènech et al. (2002); Weber (2003) and numerous other papers for their accompanying instructions.

between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$'s of the average of all the numbers provided by the participants (including your own). For example, if you choose 10 and your opponent chooses 20, then you will be the winner, as 10 is closer to $15(\text{average}) * 2/3$. Or, if you choose 30 and your opponent chooses 20, your opponent will be the winner, as 20 is closer to $25(\text{average}) * 2/3$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. Please pick one number between 0 and 100 inclusive.

User for 11-person ($p=1/2$)

Including yourself, there are 11 participants in total. The other participant you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $1/2$ of the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 10 will be the winner, as 10 is closer to $20(\text{average}) * 1/2$. Or, if you choose 25 and your opponents choose 90, 100, 50, 20, 60, 80, 5, 70, 10, 40, then you will be the winner, as 25 is closer to $50(\text{average}) * 1/2$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. Please pick one number between 0 and 100 inclusive.

User for Baseline: 11-person ($p=2/3$)

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$'s of the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then you will be the winner, as 14 is closer to $20(\text{average}) * 2/3$. Or, if you choose 25 and your opponents choose 90, 100, 50, 20, 60, 80, 5, 70, 10, 40, then the opponent with 40 will be the winner, as 40 is closer to $50(\text{average}) * 2/3$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. Please pick one number between 0 and 100 inclusive.

User for 11-person ($p=4/3$)

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $4/3$'s of the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 30 will be the winner, as 30 is closer to $20(\text{average}) \cdot 4/3$. Or, if you choose 25 and your opponents choose 90, 100, 50, 20, 60, 80, 5, 70, 10, 40, then the opponent with 70 will be the winner, as 70 is closer to $50(\text{average}) \cdot 4/3$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. Please pick one number between 0 and 100 inclusive.

User for n-person ($p=2/3$)

The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$'s of the average of all the numbers provided by the participants (including your own). For example, suppose there are 11 participants (including yourself). In this situation, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then you will be the winner, as 14 is closer to $20(\text{average}) \cdot 2/3$. Or, if you choose 25 and your opponents choose 5, 10, 20, 40, 50, 60, 70, 80, 90, 100, then the opponent with 40 will be the winner, as 40 is closer to $50(\text{average}) \cdot 2/3$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. Please pick one number between 0 and 100 inclusive.

User for Median

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $2/3$'s of the median of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 11 will be the winner, as 11 is closer to $20(\text{median}) \cdot 2/3$. Or, if you choose 12 and your opponents choose 90, 100, 50, 20, 60, 80, 5, 70, 10, 40, then the opponent with 40 will be the winner, as 40 is closer to $50(\text{median}) \cdot 2/3$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start

this game. Please pick one number between 0 and 100 inclusive.

B.3.3 Repeated pBCG

For the repeated pBCG in which each of the LLMs play within a group of 10 other LLM subjects of the same model, the system prompt is

System

I want you to act as a human decision maker, You will be participating in 10 rounds of a game. All rounds are independent of each other meaning decisions made in each round do not affect subsequent rounds. You should use your best judgment to come up with responses that you like most. Always write your final answer in [].

For the first round, the user prompt is

User

Including yourself, there are 11 participants in total. The other participants you will be playing with are human decision makers. For each round, you will play with the same set of participants. All participants will be asked to pick a number between 0 and 100 inclusive. The winner will be the one choosing the number closest to $\frac{4}{3}$'s of the average of all the numbers provided by the participants (including your own). For example, if you choose 14 and your opponents choose 10, 22, 23, 30, 11, 16, 37, 18, 19, 20, then the opponent with 30 will be the winner, as 30 is closer to $20(\text{average}) * \frac{4}{3}$. Or, if you choose 25 and your opponents choose 90, 100, 50, 20, 60, 80, 5, 70, 10, 40, then the opponent with 70 will be the winner, as 70 is closer to $50(\text{average}) * \frac{4}{3}$. In case of multiple winners, only one participant will be randomly chosen to win. Now let's start this game. After each round, all participants will be told the average, and the target ($\frac{4}{3} * \text{average}$). You will also be privately informed of whether you won or lost at the end of each round. Round 1. Please pick one number between 0 and 100 inclusive.^a

^aThe same structure for the prompt is used for $p=2/3$, replacing the examples with the $p=2/3$ examples used as in the 1-shot pBCG case.

We keep the whole history for each LLM subject in subsequent rounds by adding to the vector of messages (i.e., stacking messages) as the input to the API. After the user prompt from the first round, the answer given by the LLM is added to the messages in the assistant

role.⁴⁵ After this, as a user role message, we prompt with

User

In the previous round, the average was **average** and the target was **average** × *p*. You **won /lost** in the previous round.

Note that we host 11 LLM agents from the same LLM model. The value of **average** is computed as the mean of the responses generated independently by these 11 agents in the previous round. The target and **won/lost** outcome are then calculated based on these responses. It is worth mentioning that each agent maintains its own unique interaction history. We then add another user role message with the prompt

User

Round 2. Please pick one number between 0 and 100 inclusive.

We repeat this process of messages, keeping the entire history of messages for each LLM subject until round 10. Note that the history given to each LLM subject is specific to that LLM subject; no other information is shared with other LLM subjects beyond the average and the target. In this way, the system message and the rules of the game are always contained in the first 2 messages of the history, as well as the entire history of choices that each LLM subject makes, with accompanying feedback, specific to each LLM subject.

B.3.4 GG

The instructions come directly from Costa-Gomes and Crawford (2006), with minimal modifications. The system prompt is

System

I want you to act as a human decision maker, You will be participating in 16 rounds of a game. All rounds are independent of each other meaning decisions made in each round do not affect subsequent rounds. You should use your best judgment to come up with responses that you like most. Always write your final answer in [].

GG. For the first round, the user prompt is

⁴⁵It is trivial to extract the actual response given the system prompt requests that the final answer is given in square brackets [].

User

In each round, you will be matched with one of the other participants, a new one in each round. You will not know which of the other participants you are matched with, and the other participants are human decision makers. Each round concerns a decision situation in which you and another person we call 's/he' (which will refer to a new person each round) separately and independently make decisions called GUESSES. Together, your and her/his guesses determine the numbers of POINTS that you and s/he earn in a round, which may be different. To choose your guesses, it may help you to understand how your and her/his guesses will determine the numbers of points that you and s/he earn in the decision situations. In each decision situation, each person has her/his own TARGET, LOWER LIMIT and UPPER LIMIT. These targets and limits may be different for you and her/him, and they may change from round to round. Otherwise, the decision situations are identical in all 16 rounds. Your and her/his targets, lower limits, and upper limits will be known to you and her/him every round. Both you and s/he will receive the same instructions and have the same information about the decision situations and the same access to your and her/his targets and limits. You (respectively, s/he) can choose your (her/his) guesses only within your (her/his) given limits for each round as explained below. After submitting guesses, you earn whichever is larger, either 0 points or 200 points minus the distance between YOUR guess and the product of YOUR target times HER/HIS guess, PLUS whichever is larger, either 0 points or 100 points minus one-tenth (1/10th) the distance between YOUR guess and the product of YOUR target times HER/HIS guess. S/he earns whichever is larger, either 0 points or 200 points minus the distance between (1)HER/HIS guess and (2)the product of HER/HIS target times YOUR guess, PLUS whichever is larger, either 0 points or 100 points minus one-tenth (1/10th) the distance between (3)HER/HIS guess and (4)the product of HER/HIS target times YOUR guess. That is, $\text{Points} = \max(0, 200 - \text{distance}) + \max(0, 100 - 0.1 * \text{distance})$, where distance is the absolute difference between Your guess and your Target*(His/Her Guess). This way of determining the number of points that you and s/he earn makes the number you earn larger, the closer your guess is to your target times her/his guess; and it makes the number s/he earns larger, the closer her/his guess is to her/his target times your guess. Only the distance matters, NOT whether the difference is positive or negative. You earn the same number of points when your guess is too high by a given amount as when it is too low by the same amount. To pass the Understanding Test and TO BE ABLE to participate in the experiment, it is important to understand how your (respectively, her/his) original guesses should be chosen to stay within your (her/his) limits. This will be done as follows. If your original guess is below your lower limit, then your guess should be adjusted UP to your LOWER limit; and if your original guess is above your upper limit, then your guess should be adjusted DOWN to

your UPPER limit. If, for example, your lower limit is 400 and you original guess 300, then your adjusted guess is 400. If your upper limit is original 600 and your guess 900, then your adjusted guess is 600. Her/his guesses are adjusted up or down to her/his lower or upper limits in the same way, except that her/his limits may be different. Remember you and s/he can choose any number within your and her/his limits, respectively. We are now starting a new round of the game. Round 1: For this round your lower limit is $*LL_{you}^{Round1}*$, your upper limit is $*UL_{you}^{Round1}*$, and your target is $*target_{you}^{Round1}*$. Her/His lower limit is $*LL_{his/her}^{Round1}*$, her/his upper limit is $*UL_{his/her}^{Round1}*$, and her/his target is $*target_{his/her}^{Round1}*$. Please choose a number between your lower limit and upper limit.

Similar to the repeated pBCG, we keep the whole history for each LLM subject in subsequent rounds by adding to the vector of messages (i.e., stacking messages) as the input to the API.⁴⁶ After the user prompt from the first round, the answer given by the LLM is added to the messages in the assistant role. Different from the pBCG, and consistent with Costa-Gomes and Crawford (2006), we do not provide feedback to avoid repeated game effects, experience-based learning, and experimentation. Therefore, after the assistant message, as a user role message, we prompt with the next round.

User

We are now starting a new round of the game. Round 2: For this round your lower limit is $*LL_{you}^{Round2}*$, your upper limit is $*UL_{you}^{Round2}*$, and your target is $*target_{you}^{Round2}*$. Her/His lower limit is $*LL_{his/her}^{Round2}*$, her/his upper limit is $*UL_{his/her}^{Round2}*$, and her/his target is $*target_{his/her}^{Round2}*$. Please choose a number between your lower limit and upper limit.

We repeat this process of messages, keeping the entire history of its own messages for each LLM subject until all 16 rounds are completed. Note that the history given to each LLM subject is specific to that LLM subject; no other information is shared with other LLM subjects beyond their history. In this way, the system message, and the rules of the game are always contained in the first 2 messages of the history, as well as the entire history of choices that each LLM subject makes.

B.3.5 MRG

The instructions come directly from Arad and Rubinstein (2012), with minimal modifications. Similarly, we avoid monetary payoffs as rewards to avoid issues with incentives. We also provide examples in the instructions as LLMs had better answers to the best responses

⁴⁶We use the term “round” in our prompts to match the original instructions in Costa-Gomes and Crawford (2006). However, a round does not refer to repeated play of the same game. Instead, each round presents a different one-shot GG task with distinct parameters.

understand questions (as this helps LLMs understand what kind of calculations are involved). The system prompt is

System

I want you to act as a human decision maker, You will be participating in 1 round of a game. You should use your best judgment to come up with responses that you like most. Always write your final answer in [].

This is identical for both MRG and MRG (Game 3). The following prompts include all the details for MRG and MRG (Game 3), specified by the user role.

User for MRG

You are randomly matched to play a game against a human decision maker. In the game, each of you chooses an integer between 11 and 20. Each participant will receive points in the amount they have chosen. A participant will receive an additional 20 points if they choose a number exactly one less chosen by the other player. For example, if the other participant chooses X, you will get $(X-1)+20$ points if you choose X-1 points. For example, if your opponent chooses 14, you will get 33 points if you choose 13. Or, if the other participant chooses 11, you will get 20 points if you choose 20. You will receive your points without knowing against whom you played. Now let's start this game. What number do you choose?

User for MRG (Game 3)

You are randomly matched to play a game against a human decision maker. In the game, each of you chooses an integer between 11 and 20. If a participant chooses the number 20, they will receive 20 points (regardless of the other player's choice). Otherwise, if a participant chooses any other number in this range, they will receive three points less than in the case where they choose 20. However, they will receive an additional amount of 20 points if they choose a number that is lower by exactly one than the number chosen by the other player. For example, if the other participant chooses X, you will get $17+20$ points if you choose X-1. For example, if your opponent chooses 14, you will get $17+20$ points if you choose 13. Or, if the other participant chooses 11 points, you will get 20 points if you choose 20. You will receive your points without knowing against whom you played. Now let's start this game. What number do you choose?

B.4 Estimation Details

B.4.1 pBCG

From the theoretical predictions, we apply standard practice maximum likelihood estimation (MLE) to estimate parameters for both the level- k model and the CH model. For the level- k model, we estimate the proportion of responses that align with the L_0 , L_1 , L_2 , L_3 , L_4 , and L_∞ (Nash) predictions. L_0 also accommodates behavior which is not precisely predicted by the level- k (or CH model) for levels (steps) higher than L_1 (step-1), i.e., the L_0 type captures all responses from a type who chooses uniformly randomly from the integers 0 to 100, but is not in the proximity of any prediction from any other higher type. This allows us to accommodate behavior which is not in the neighborhood of the predictions of any of the level- k types (up to a finite k).⁴⁷ For each type, we assume that a choice is the best guess limited by their type, plus an independent and identically distributed (i.i.d.) error term. Specifically, assuming the error term is i.i.d. binomially distributed, the observed choice of a level- k type by an individual, i , is given by $Y_i = l_k + \epsilon_i$, with $\epsilon_i \sim \mathcal{B}(\alpha, \beta)$ such that $E[\epsilon_i] = 0$, and $\text{Var}[\epsilon_i] = \sigma^2$. With $\epsilon_i \in \mathbb{R}$ and $\epsilon_i = Y_i - l_k$, this allows us to define a density for each of these types:

$$h_k(\epsilon_i) = \begin{cases} \binom{\alpha}{\epsilon_i} \beta^{\epsilon_i} (1 - \beta)^{\alpha - \epsilon_i} & k \in \{1, \dots, K, \infty\}, \\ \frac{1}{101} & k = 0. \end{cases}$$

Let $\mathcal{K}$ denote the space of types, i.e., $\mathcal{K} = \{0, \dots, K, \infty\}$. Define proportions of types by $\mathcal{F} = \{f_k\}_{k \in \mathcal{K}}$, where f_k is the proportion of level- k types. Defining our complete parameter space as $\Theta = (\mathcal{F}, \alpha, \beta)$ with α, β being the nuisance parameters, the log-likelihood function for a random sample of guesses, $\mathbf{Y} = Y_{i \in \{1, \dots, N\}}$ is:

$$\log \mathcal{L}(\Theta | \mathbf{Y}) = \sum_{i=1}^n \ln \left(\sum_{k \in \mathcal{K}} f_k \cdot h_k(\epsilon_i) \right).$$

The purpose of assuming a binomially distributed random error is to allow for discrete-valued errors, in essence, we use a *discretized* normal distribution centered around 0 (with its limit being a normal distribution). We estimate the level- k model with 6 levels of reasoning

⁴⁷The goal of MLE is to find the parameter values that maximize the product of the individual likelihoods, based on a (probabilistic) matching of the theoretical predictions and the observed strategies. However, an action predicted to occur with zero probability results in a likelihood contribution of zero, rendering the overall likelihood function zero. Such an event, even if it occurs for only one observation, can obscure the predictive accuracy of any model. To avoid the zero-likelihood problem, we introduce a type that randomizes across all possible strategies, ensuring that no action is chosen with zero probability, as described in Moffatt (2015) and footnote 2 of Camerer, Ho and Chong (2004).

by setting $K = 4$ with the best guess for the L_∞ type being the Nash equilibrium, the L_0 type being a uniform randomizer, and the remaining best guesses being determined by the theoretical predictions above.

For the CH model, τ is estimated, which represents the mean (and variance) of the frequency distribution of step- k types. τ parameterizes the proportion of responses that align with step- k (for $k \in \{0, \dots, K\}$) and step- ∞ predictions. The structure of the log-likelihood function is identical to the one used to estimate the level- k model, with choices also modeled as noisy best guesses in the same way as above. The differences only lie in 1) the proportions of types are now determined by a Poisson distribution, i.e., the distribution of types is given by $f(k; \tau)$, and 2) the best guesses are made by assuming a distribution over lower step- k , determined by the conditional Poisson distribution $\{f_k(j; \tau)\}_{j=0,1,\dots,k-1}$, i.e., with $\epsilon_i \in \mathbb{R}$ and $Y_i = s_k(\tau) + \epsilon_i$. This structure is identical to the level- k model except we replace the level- k best guess, l_k , with the CH model best guess, $s_k(\tau)$.

Let $\mathcal{K}$ denote the space of types, i.e., $\mathcal{K} = \{0, \dots, K, \infty\}$. Define proportions of types by $\{f(k; \tau) | k \in \mathcal{K}\}$, where $f(k; \tau)$ is the proportion of step- k types, given τ .⁴⁸ The complete parameter space is $\Theta = (\tau, \alpha, \beta)$ with α, β being the nuisance parameters, hence, the log-likelihood function for a random sample of guesses, $\mathbf{Y} = Y_{i \in \{1, \dots, N\}}$ is:

$$\log \mathcal{L}(\Theta | \mathbf{Y}) = \sum_{i=1}^n \ln \left(\sum_{k \in \mathcal{K}} f(k; \tau) \cdot h_k(\epsilon_i) \right).$$

Although the log-likelihood function is of the same structure, estimation is vastly different as the CH model is fully determined by the value of τ , i.e., finding the value of τ that maximizes the log-likelihood. For consistency, we estimate using the same number of step- k type players as there are level- k type players.⁴⁹

B.4.2 GG

Following the theoretical predictions, we use similar MLE techniques to estimate parameters for both the level- k model and the CH model. The notable difference here is that we collected data sequentially for the 16 rounds of the GG. For each LLM, we collected this set of responses 100 times (i.e., for each LLM, there are 100 LLM subjects playing 16 rounds of the GG sequentially). With these data, for the level- k model, we are able to estimate the proportion of responses that align with the L_0 , L_1 , L_2 , L_3 , L_4 , and L_∞ (Nash) predictions on an individual-response level. We do this because each round has its own different set of predictions (level- k ,

⁴⁸ f_0 is still parameterized by τ by the law of total probability.

⁴⁹ Note that α and β are simply noise parameters for the estimation in the binomial distribution, with no mapping to the theoretical models, as such, they are nuisance parameters.

CH, Nash), so we need not base the estimation on aggregate responses, as done in the pBCG.⁵⁰

The observed (noisy) choice of a level- k type by an individual, i , for each round, r , is given by $Y_i^r = l_k^r + \epsilon_i^r$, with $\epsilon_i^r \sim \mathcal{B}(\alpha^r, \beta^r)$ such that $E[\epsilon_i^r] = 0$, and $\text{Var}[\epsilon_i^r] = (\sigma^r)^2$. With $\epsilon_i^r \in \mathbb{R}$ and $\epsilon_i^r = Y_i^r - l_k^r$, this allows us to define a density for each of these types:

$$h_k^r(\epsilon_i) = \begin{cases} \binom{\alpha^r}{\epsilon_i^r} \beta^{\epsilon_i^r} (1 - \beta)^{\alpha^r - \epsilon_i^r} & k \in \{1, \dots, K, \infty\}, \\ \frac{1}{b_i^r - a_i^r} & k = 0. \end{cases}$$

Depending on the parameters of the GG, the level of reasoning required to reach the L_∞ prediction can be fewer than $K(> 1)$. In these cases, we assume the response is only consistent with the L_∞ (and L_0) type. Formally, for any $k \in \{1, \dots, K\}$ and $l_k^r = l_\infty^r$, set $h_j^r(\epsilon_i) = 0$, for $j \leq k$, with $h_\infty^r(\epsilon_i)$ and $h_0^r(\epsilon_i)$ defined as above.⁵¹

Let $\mathcal{K}$ denote the space of types, i.e., $\mathcal{K} = \{0, \dots, K, \infty\}$. Define proportions of types by $\mathcal{F} = \{f_k\}_{k \in \mathcal{K}}$, where f_k is the proportion of level- k types. Defining our complete parameter space as $\Theta = (\mathcal{F}, \alpha, \beta)$ with α, β collecting nuisance parameters in vectors, the log-likelihood function for a random sample of guesses, $\mathbf{Y}_i = Y_{i,r \in \{1, \dots, 16\}}$ is:

$$\log \mathcal{L}(\Theta | \mathbf{Y}_i) = \sum_{r=1}^{16} \ln \left(\sum_{k \in \mathcal{K}} f_k \cdot h_k^r(\epsilon_i) \right).$$

Notice this is virtually identical to the MLE used in the pBCG except we use data collected for all rounds per LLM subject to estimate the level- k model, rather than aggregating over all the data collected. In this way, we can estimate the level- k model for each LLM subject. As we estimate up to L_4 , including L_0 and L_∞ , we set $K = 4$.

For the CH model, τ is estimated in an analogous way. Define proportions of types in each round by $\{f(k; \tau) | k \in \{0, \dots, K, \infty\}\}$, where $f(k; \tau)$ is the proportion of step- k types, given τ , with $\epsilon_i^r \in \mathbb{R}$ and $Y_i^r = s_k^r(\tau) + \epsilon_i^r$. As before we impose that, for any $k \in \{1, \dots, K\}$ and $l_k^r = l_\infty^r$, set $h_j^r(\epsilon_i) = 0$, for $j \leq k$, with $h_\infty^r(\epsilon_i)$ and $h_0^r(\epsilon_i)$ defined as above. The complete parameter space is $\Theta = (\tau, \alpha, \beta)$ with α, β collecting nuisance parameters in vectors, hence, the log-likelihood function for a random sample of guesses, $\mathbf{Y}_i = Y_{i,r \in \{1, \dots, 16\}}$ is:

⁵⁰Naturally, another way to approach the estimation is to aggregate the data for each round, and estimate the model for each of the 16 rounds of the GG, as if each round is their own one-shot game. However, this does not make use of the sequential nature in which the GG is played in Costa-Gomes and Crawford (2006), which we attempt to stay as close to as possible, in terms of experimental design.

⁵¹While the majority of the mass goes to the L_∞ proportion when the response is close to the L_∞ prediction, we still allow for random behavior.

$$\log \mathcal{L}(\Theta | \mathbf{Y}_i) = \sum_{r=1}^{16} \ln \left(\sum_{k \in \mathcal{K}} f(k; \tau) \cdot h_k^r(\epsilon_i) \right).$$

Again, the log-likelihood function is of the same structure, but estimation differs because the CH model is fully determined by τ , separately for each LLM subject. For consistency, we estimate using the same number of step- k type players as there are level- k type players.⁵²

B.4.3 MRG

Estimation. Similarly, from the theoretical predictions, we apply MLE to estimate parameters for both the level- k model and the CH model analogously to what was done for the pBCG and GG. We keep the same number of levels/steps as the pBCG but replace L_∞ with L_4 , as there is no pure Nash equilibrium. We still accommodate behavior which is not predicted by the level- k or CH model by including a purely random type. However, as L_0 behavior can be very distinct from purely random behavior, we specifically distinguish between these two types. Another reason for doing this is to allow for the uniform randomization choice (15), and lower choices, to be considered as random behavior, rather than a very high stage of reasoning.⁵³ Estimation is done separately for the MRG and MRG (Game 3).

Given the finite nature of the actions in MRG and MRG (Game 3), applying MLE to estimate the parameters for the level- k model and the CH model is simple and standard. For the level- k model, we estimate the proportion of levels that are consistent with purely random behavior, L_0 , L_1 , L_2 , L_3 , and L_4 predictions. Let $\mathcal{K}$ denote the space of types, i.e., $\mathcal{K} = \{\text{random}, 0, 1, \dots, K\}$. We define the density for each type as follows:

$$h_k(Y_i) = \begin{cases} \mathbb{1}[Y_i = l_k] & k \in \{0, \dots, K\}, \\ \frac{1}{10} & k = \{\text{random}\}. \end{cases}$$

where $\mathbb{1}[Y_i = l_k]$ is an indicator function which takes 1 when $Y_i = l_k$ and 0 otherwise. Define proportions of types by $\mathcal{F} = \{f_k\}_{k \in \mathcal{K}}$, where f_k is the proportion of each type. We set $K = 4$ to estimate 6 levels of reasoning (including L_0 , and purely random type). The parameter space is simply $\Theta = \mathcal{F}$ and the log-likelihood function for a random sample of

⁵²Note that α and β are again (vectors of) noise parameters for the estimation in the binomial distribution, with no mapping to the theoretical models.

⁵³This is consistent with the empirical results in Arad and Rubinstein (2012) with virtually no subject choosing below 15.

guesses, $\mathbf{Y} = Y_{i \in \{1, \dots, N\}}$ is:

$$\log \mathcal{L}(\Theta | \mathbf{Y}) = \sum_{i=1}^n \ln \left(\sum_{k \in \mathcal{K}} f_k \cdot h_k(Y_i) \right).$$

For the CH model, we estimate τ to parameterize the proportion of responses that align with step- k , for $k \in \{\text{random}, 0, 1, \dots, K\}$, to be consistent with the level- k model. As before, the proportions of types are determined by a Poisson distribution with best responses as given above in the theoretical predictions. Define proportions of types by $\{f(k; \tau) | k \in \mathcal{K}\}$ where $f(k; \tau)$ is the proportion of step- k types, given τ . The parameter space is simply $\Theta = \{\tau\}$ and the log-likelihood function for a random sample of guesses, $\mathbf{Y} = Y_{i \in \{1, \dots, N\}}$ is:

$$\log \mathcal{L}(\Theta | \mathbf{Y}) = \sum_{i=1}^n \ln \left(\sum_{k \in \mathcal{K}} f(k; \tau) \cdot \mathbb{1}[Y_i = s_k(\tau)] \right).$$

As with the pBCG and GG, the log-likelihood function is of the same structure between models, with different estimation, given the way that the proportions are parameterized. For consistency, the same number of step- k players is used as there are level- k players.

B.5 Temperature

The temperature parameter influences outputs in terms of randomness and creativity. A lower temperature value results in more predictable and deterministic responses, in favor of words with the highest probabilities. Conversely, a higher temperature results in more diverse and novel responses by increasing the chance of less likely words. In line with studies exploring the impact of temperature settings on LLM behavior (Loya, Sinha and Futrell, 2023; Binz and Schulz, 2023; Zhu et al., 2024), every condition mentioned is also tested at 3 temperature levels. The baseline is set at a moderate temperature of 0.5 to ensure the most balanced approach. We have higher (0.75) and lower (0.25) temperature level conditions to examine how variations in randomness influence (observed) strategic sophistication. The exception is with GPT-o1 as at the time of collecting the data (October 2024) varying the temperature parameter was not available. Given the results of the baseline (for all games) and as we only use GPT-o1, Cl-3S, Cl-4ST and Ge-2T for the GG, we did not run the GG with different temperatures. Given that the level- k model results show very little variation across temperature, the CH model estimates also do not vary much from baseline temperature either.

B.5.1 pBCG

For all conditions (baseline and alternate), we collected data varying the temperature to determine whether there is any impact on LLM responses. Overall, the results remained largely consistent with those observed at the baseline temperature. The only differences of note occurred in the higher temperature condition, in which there was an increase in incoherent answers, and a slight rise in the variation of responses.⁵⁴ Analogously, for lower temperatures, the responses became more concentrated at the mode, as expected with more deterministic results. However, the estimates of either model remained relatively unchanged. This consistency across different temperatures suggests that strategic sophistication of LLMs is robust to changes in temperature. We present the results for the different temperature conditions below.

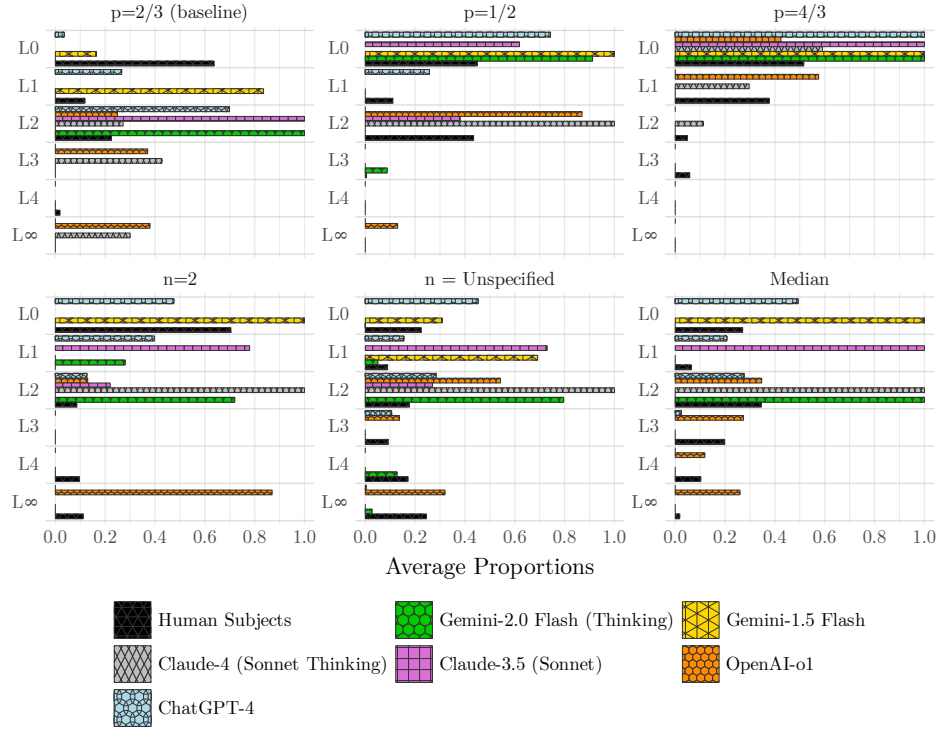


Figure B2: Estimates of the proportions of level- k types for the pBCG with high temperature.

⁵⁴We still maintain the planned sample size after dropping incoherent answers.

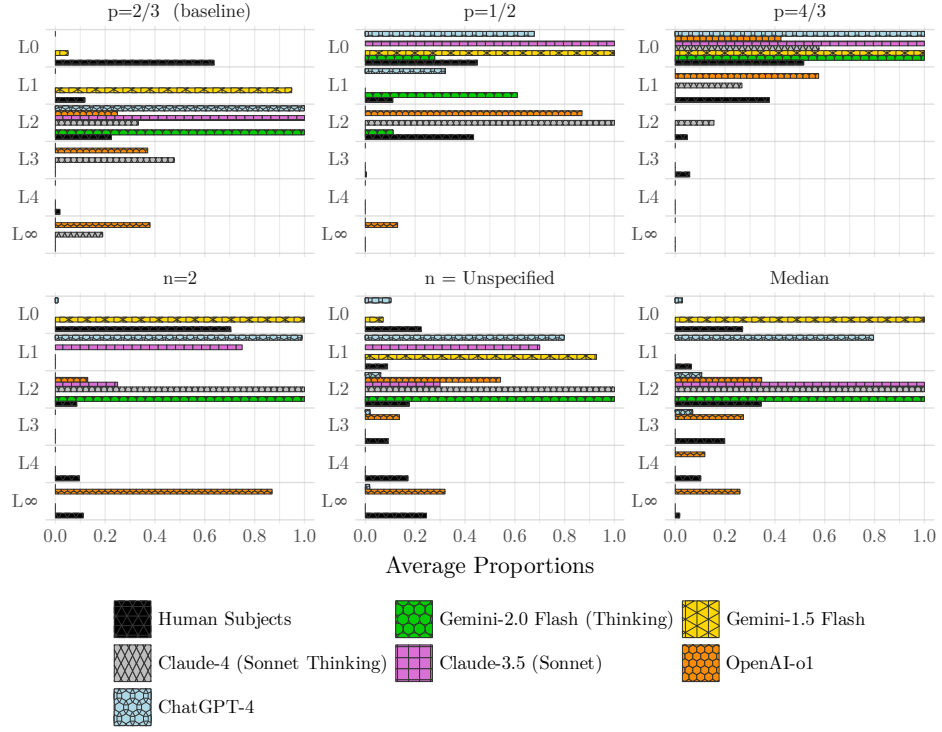


Figure B3: Estimates of the proportions of level- k types for the pBCG with low temperature.

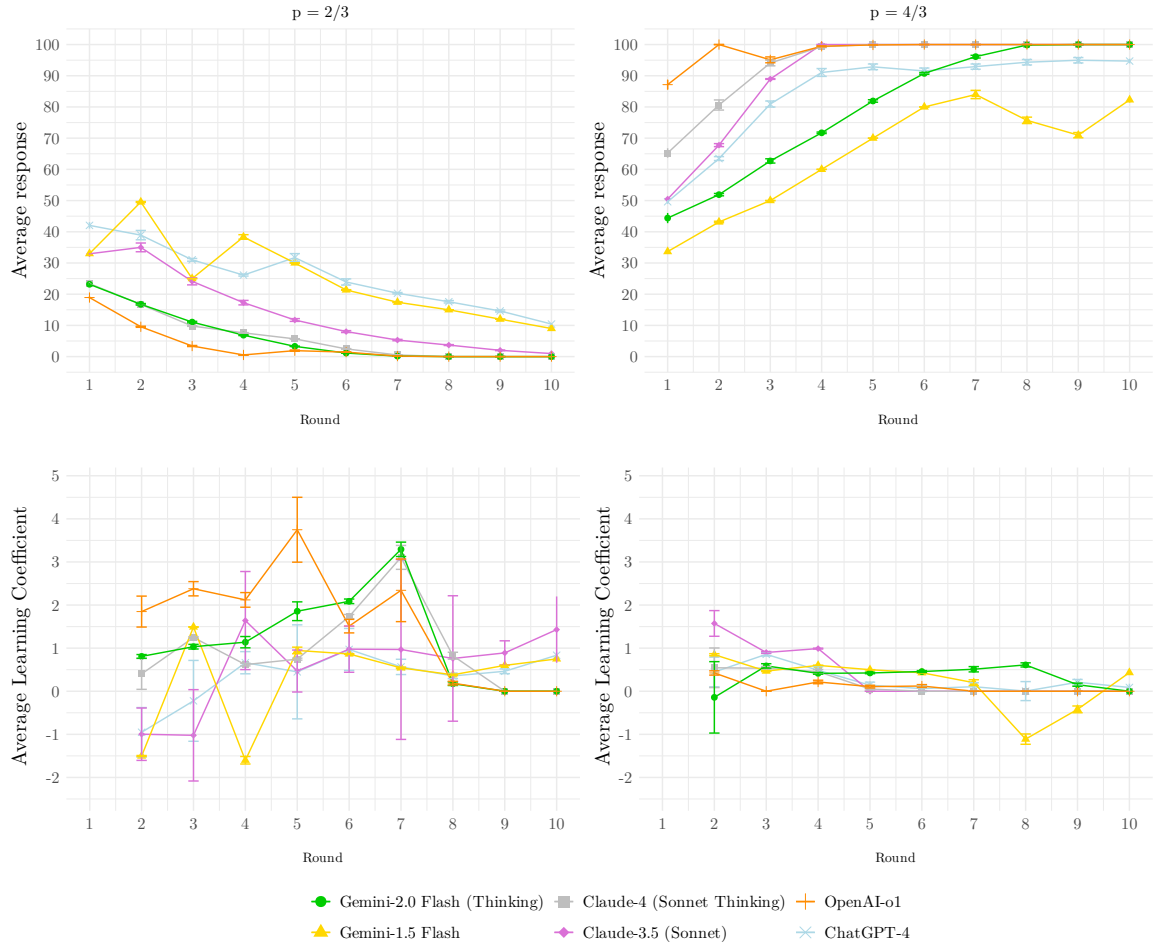


Figure B4: Time series of responses and learning coefficients for multiple rounds pBCG. Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals (high temperature).

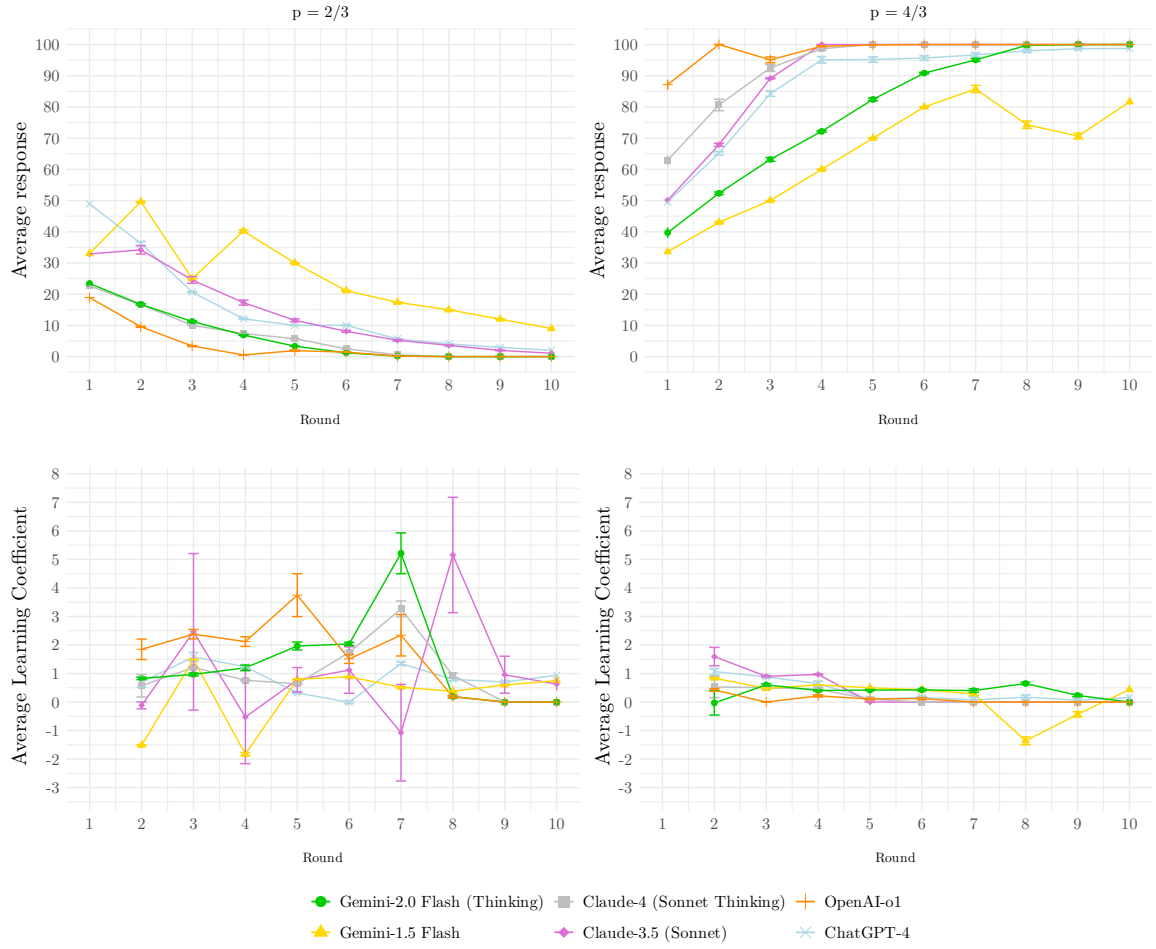


Figure B5: Time series of responses and learning coefficients for multiple rounds pBCG. Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals (low temperature).

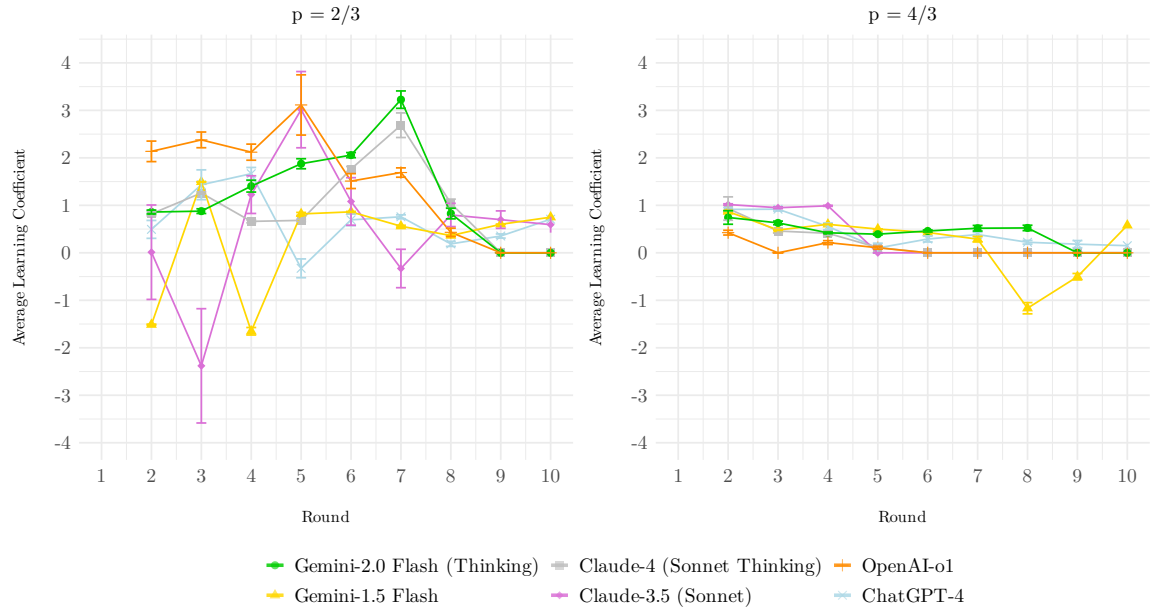


Figure B6: Time series of responses and learning coefficients for multiple rounds pBCG, excluding rounds with low errors (denominators). Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals (baseline temperature).

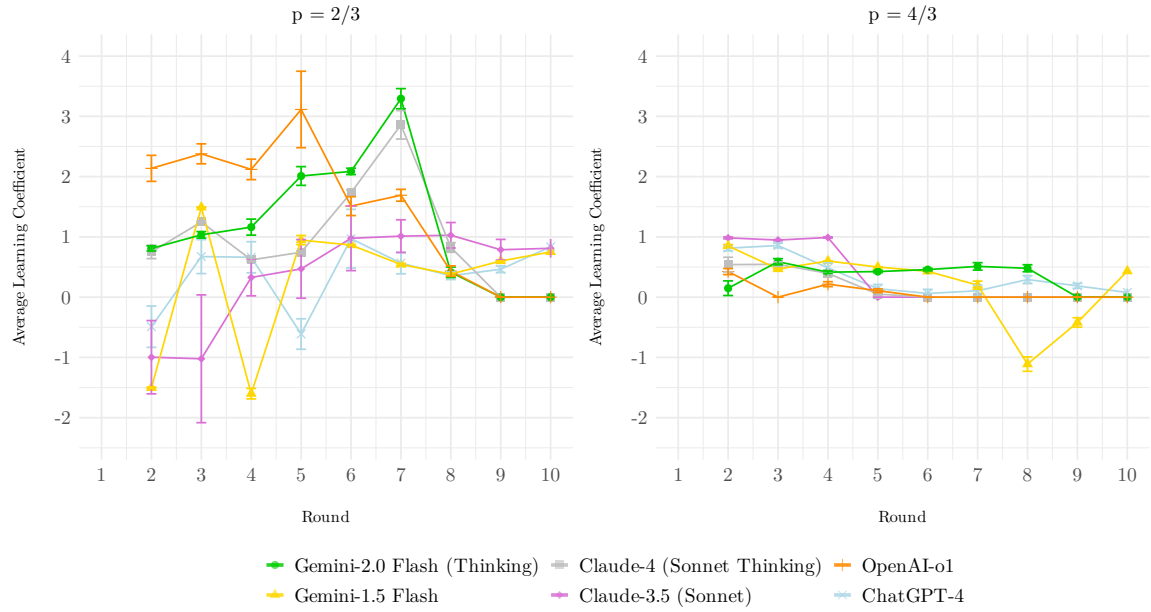


Figure B7: Time series of responses and learning coefficients for multiple rounds pBCG, excluding rounds with low errors (denominators). Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals (low temperature).

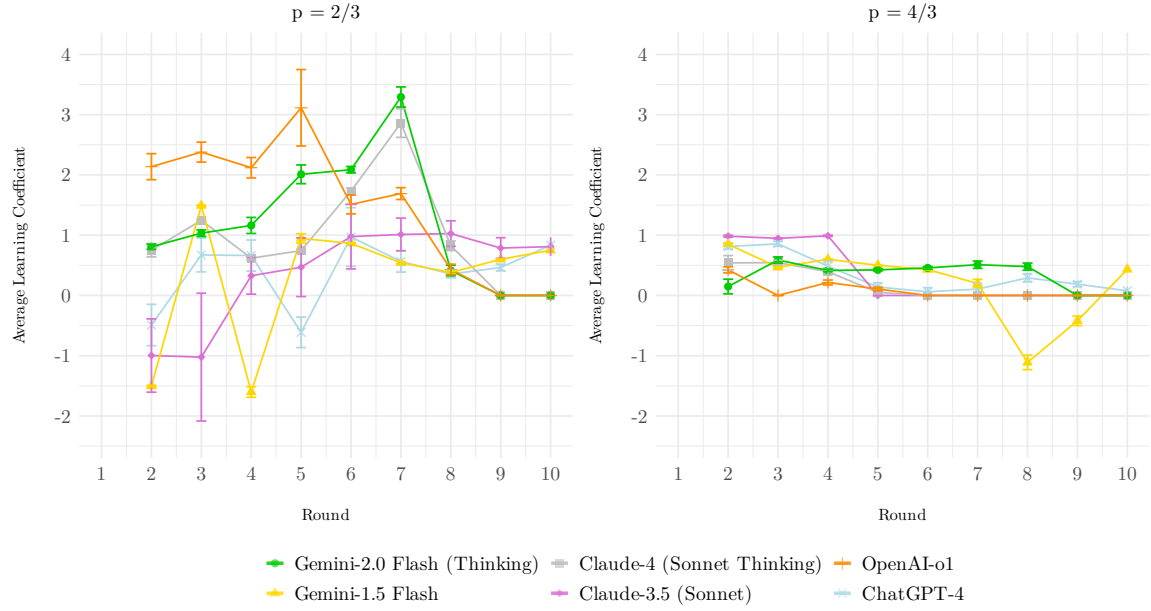


Figure B8: Time series of responses and learning coefficients for multiple rounds pBCG, excluding rounds with low errors (denominators). Each figure shows averages in each round for each LLM. Vertical bars are 95% confidence intervals (high temperature).

B.5.2 MRG

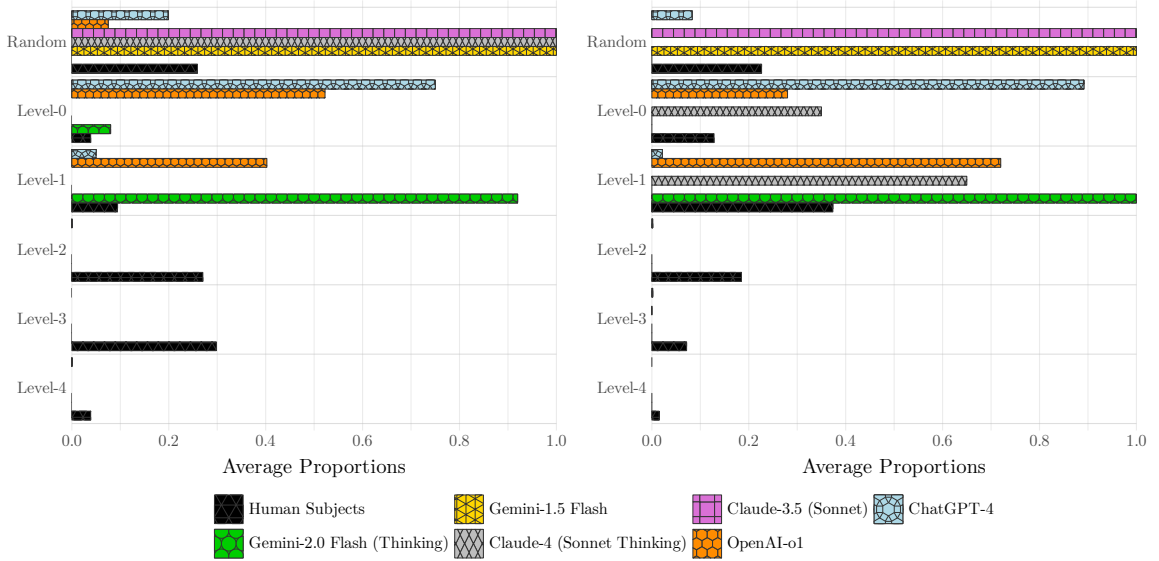


Figure B9: Estimates of the proportions of level- k types for MRG, and MRG (Game 3) with high temperature.

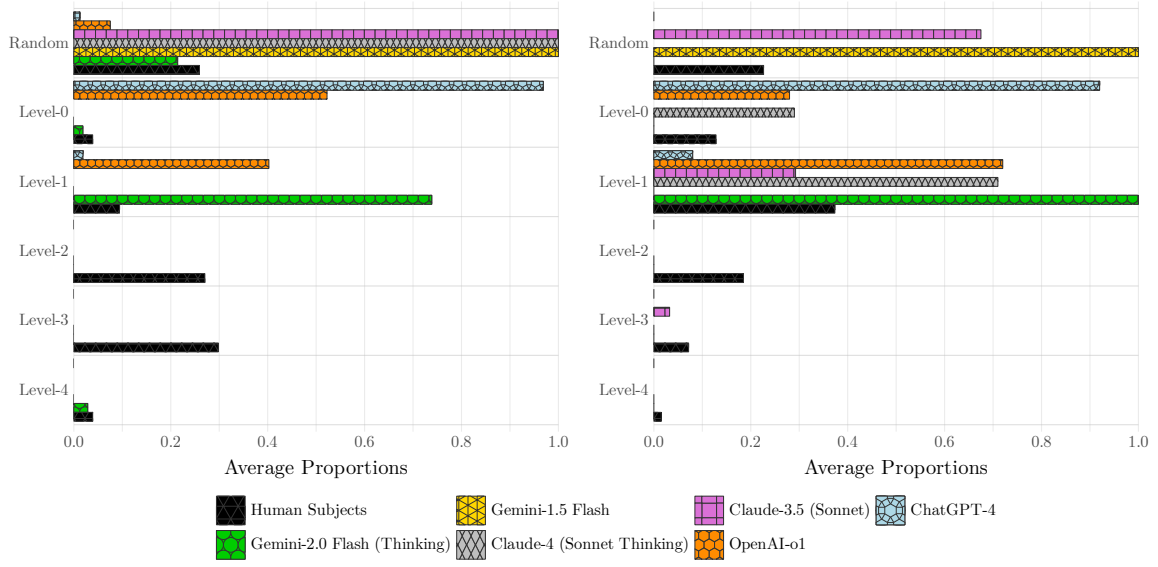


Figure B10: Estimates of the proportions of level- k types for MRG, and MRG (Game 3) with low temperature.

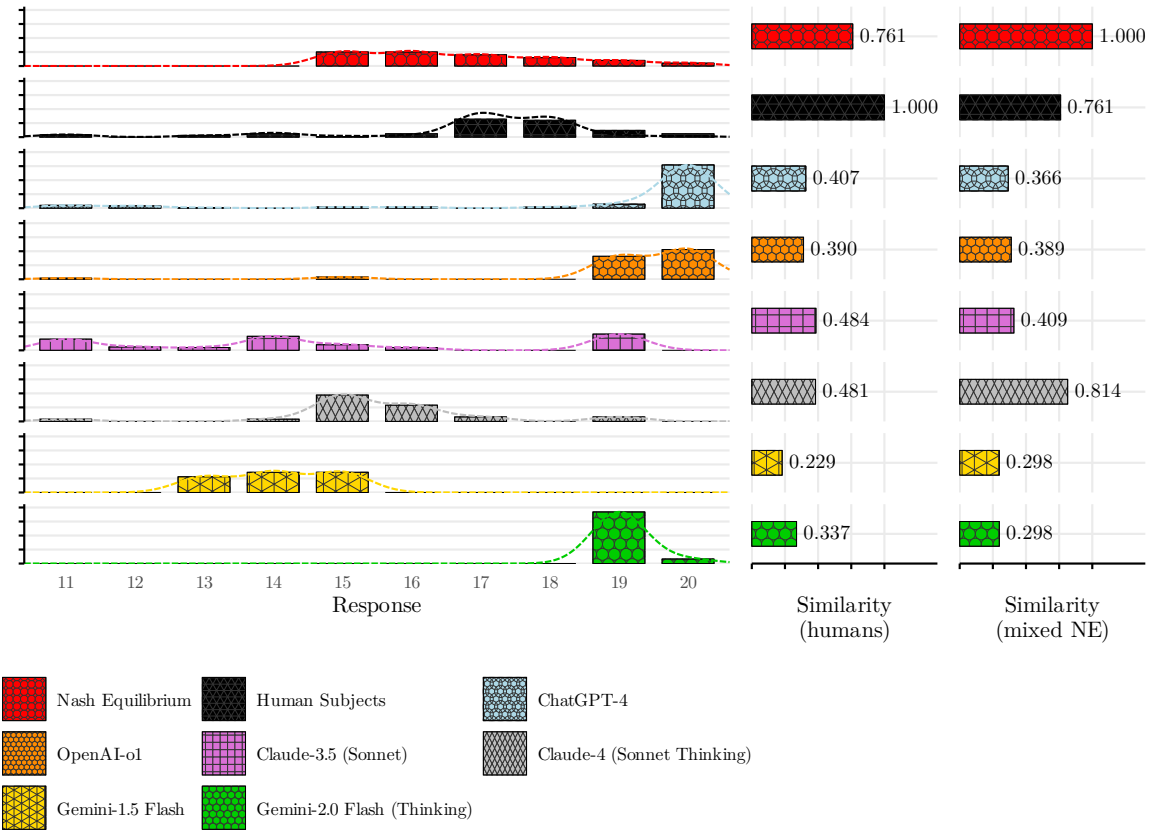


Figure B11: Histograms of responses in the MRG (left panel) alongside similarity measures (middle and right panels). The horizontal bars reflect the magnitude of similarity between the LLMs' response distributions and those of human subjects and the mixed Nash equilibrium, respectively, measured as 1 minus the Jensen-Shannon divergence. Dashed continuous density plots are overlaid on the histograms (omitted where over 98% of responses are concentrated on any single response) (high temperature).

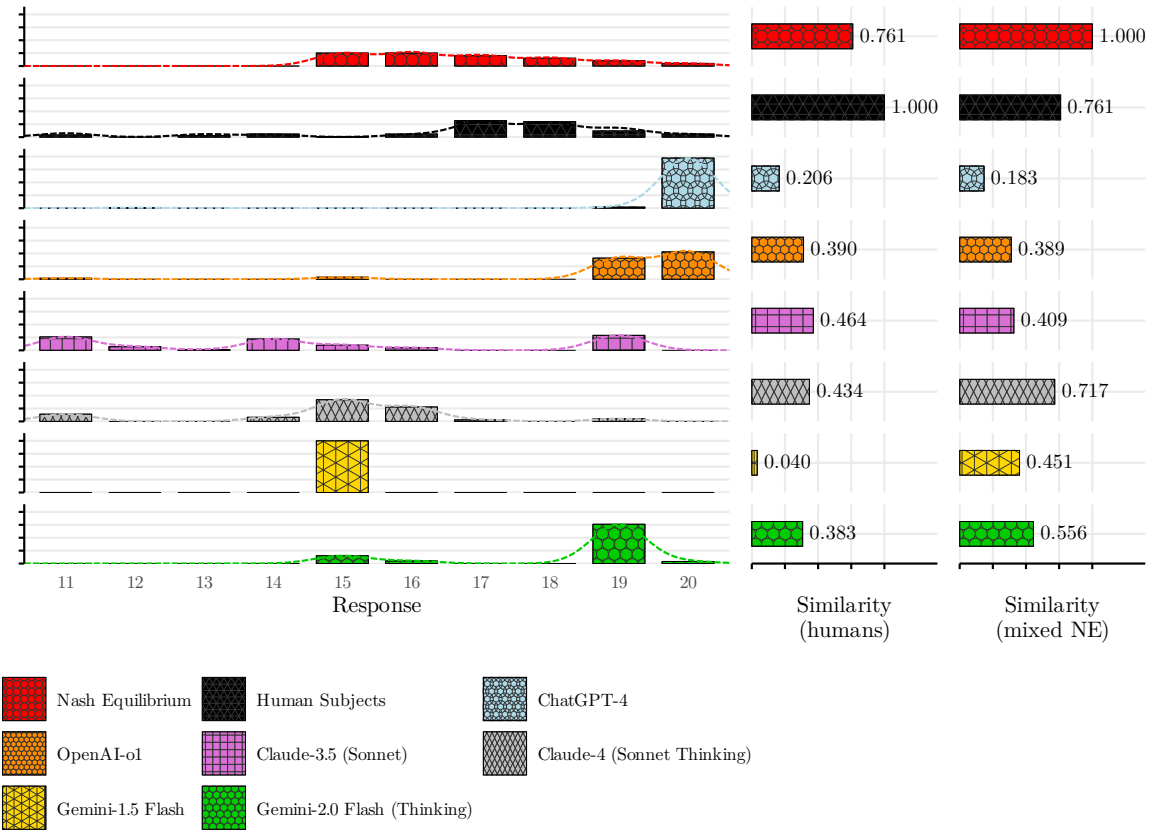


Figure B12: Histograms of responses in the MRG (left panel) alongside similarity measures (middle and right panels). The horizontal bars reflect the magnitude of similarity between the LLMs' response distributions and those of human subjects and the mixed Nash equilibrium, respectively, measured as 1 minus the Jensen-Shannon divergence. Dashed continuous density plots are overlaid on the histograms (omitted where over 98% of responses are concentrated on any single response) (low temperature).

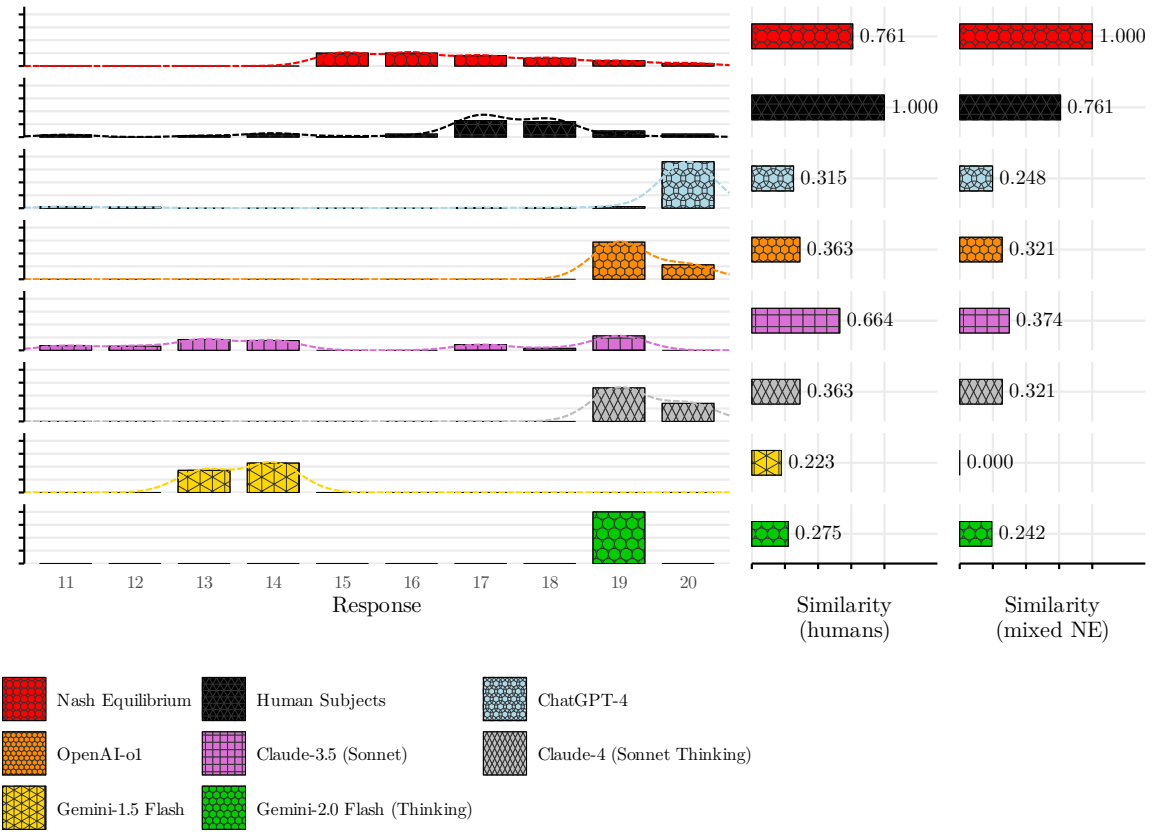


Figure B13: Histograms of responses in the MRG (left panel) alongside similarity measures (middle and right panels). The horizontal bars reflect the magnitude of similarity between the LLMs' response distributions and those of human subjects and the mixed Nash equilibrium, respectively, measured as 1 minus the Jensen-Shannon divergence. Dashed continuous density plots are overlaid on the histograms (omitted where over 98% of responses are concentrated on any single response) (high temperature).

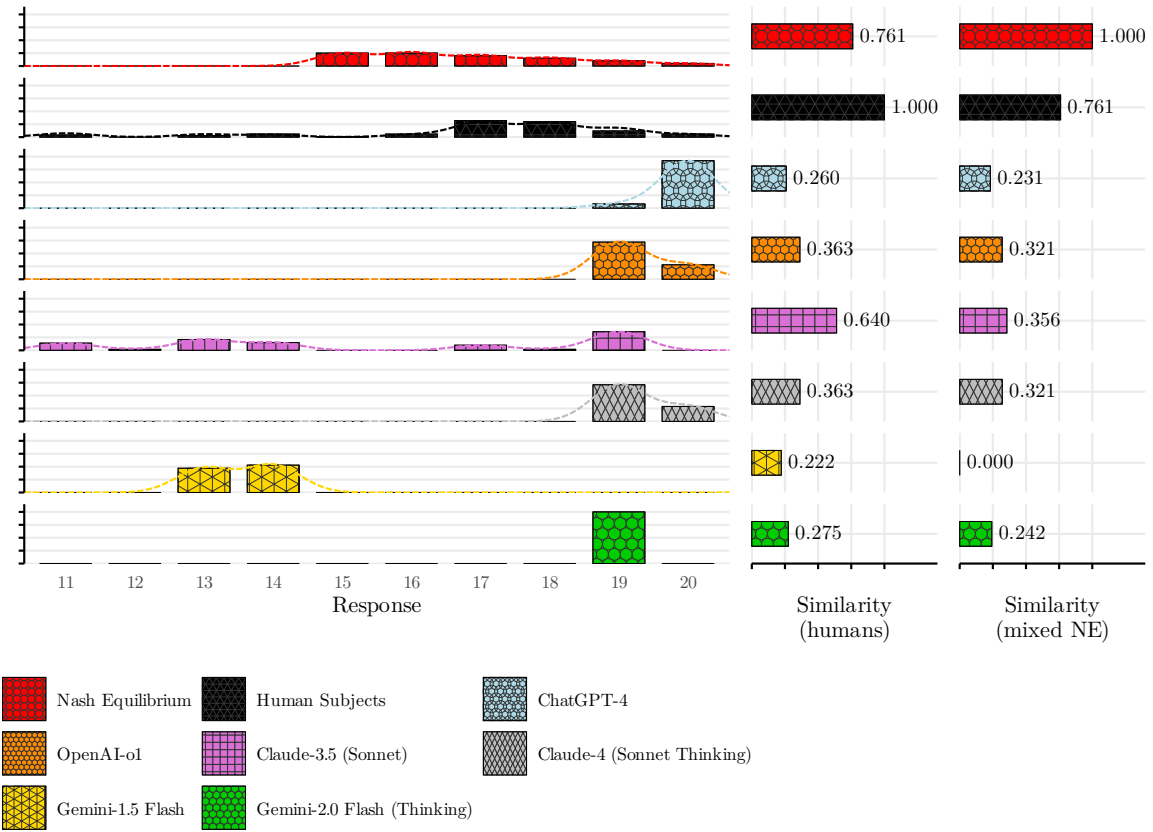


Figure B14: Histograms of responses in the MRG (left panel) alongside similarity measures (middle and right panels). The horizontal bars reflect the magnitude of similarity between the LLMs' response distributions and those of human subjects and the mixed Nash equilibrium, respectively, measured as 1 minus the Jensen-Shannon divergence. Dashed continuous density plots are overlaid on the histograms (omitted where over 98% of responses are concentrated on any single response) (low temperature).